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Nonlinear dynamics of flexible structures using corotational beam elements

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Abstract

The purpose of this thesis is to develop corotational beam elements for the nonlinear dynamic analyse of flexible beam structures. Whereas corotational beam elements in statics are well documented, the derivation of a corotational dynamic formulation is still an issue.

In the first journal paper, an efficient dynamic corotational beam formulation is proposed for 2D analysis. The idea is to adopt the same corotational kinematic description in static and dynamic parts. The main novelty is to use cubic interpolations to derive both inertia terms and internal terms in order to capture correctly all inertia effects. This new formulation is compared with two classic formulations using constant Timoshenko and constant lumped mass matrices.

In the second journal paper, several choices of parametrization and several time stepping methods are compared. To do so, four dynamic formulations are investigated. The corotational method is used to develop expressions of the internal terms, while the dynamic terms are formulated into a total Lagrangian context. Theoretical derivations as well as practical implementations are given in detail. Their numerical accuracy and computational efficiency are then compared. Moreover, four predictors and various possibilities to simplify the tangent inertia matrix are tested.

In the third journal paper, a new consistent beam formulation is developed for 3D analysis. The novelty of the formulation lies in the use of the corotational framework to derive not only the internal force vector and the tangent stiffness matrix but also the inertia force vector and the tangent dynamic matrix. Cubic interpolations are adopted to formulate both inertia and internal local terms. In the derivation of the dynamic terms, an approximation for the local rotations is introduced and a concise expression for the global inertia force vector is obtained. Four numerical examples are considered to assess the performance of the new formulation against two other ones based on linear interpolations.

Finally, in the fourth journal paper, the previous 3D corotational beam element is extended for the nonlinear dynamics of structures with thin-walled cross-section by introducing the warping deformations and the eccentricity of the shear center. This leads to additional terms in the expressions of the inertia force vector and the tangent dynamic matrix. The element has seven degrees of freedom at each node and cubic shape functions are used to interpolate local transversal displacements and axial rotations. The performance of the formulation is assessed through five examples and comparisons with Abaqus 3D-solid analyses.

Keywords: corotational method; nonlinear dynamics; large displacements; finite rotations; time stepping method; thin-walled cross-section; beam element;

Sammanfattning

Syftet med denna avhandling är att undersöka det icke linjära dynamiska beteendet av flexibla balkstrukturer med hjälp av den co roterade metoden. Co roterade balk element i statik är väl dokumenterat. Men härledningen av en dynamisk formulering är fortfarande ett olöst problem.

En effektiv dynamisk formulering föreslås för 2D analyser. Idéen är att använda samma kinematiska beskrivning för både statiska och dynamiska termer. Den viktigaste nyheten är att kubisk interpolering används för att härleda både tröghets- och interna termer, vilket möjliggör en korrekt modellering av tröghetseffekter. Den nya formuleringen jämförs med två andra klassiska formuleringar baserade på konstant diagonal och konstant Timoshenko mass matriser. Detta arbete presenteras i den första bifogade tidskriftartikeln.

Vidare diskuteras parametrisering av stora rotationer, vilket är en central fråga i utvecklingen av 3D olinjära balkelement. Huvudsyftet är att jämföra flera parametriseringar och tidsintegrationsmetoder. För att göra detta, testas fyra dynamiska formuleringar. Den co roterade metoden används för att härleda interna termer, medan de dynamiska termerna formuleras i total Lagrangian sammanhang. Teoretiska beräkningar samt praktiska implementationer ges i detalj. Numerisk noggrannhet och beräkningseffektivitet jämförs. Fyra prediktorer och olika möjligheter att förenkla tangent tröghetsmatrisen testas. Detta arbete presenteras i den andra bifogade tidskriftartikeln.

En ny konsistent dynamisk formulering föreslås för 3D-analyser. Huvudidéen är att den co roterade metoden används för att härleda både statiska och dynamiska termer. Kubisk interpolering används för att formulera både lokala interna och lokala tröghets termer. En approximation i uttrycket för de lokala rotationerna används för att förenkla de dynamiska termerna och få ett konstant uttryck för vektorn med globala tröghetskrafter. Fyra numeriska exempel testas för att jämföra den nya metoden mot två andra baserade på linjär interpolering. Detta arbete presenteras i den tredje bifogade tidskriftartikeln.

Slutligen undersöks det olinjära dynamiska beteendet av balkar med tunnväggiga tvärsnitt. Den ursprungliga formuleringen för 3D balkar utvidgas genom att introducera deformationer från välvning och ett excentriskt skjuvcentrum. Detta leder till extra termer i uttrycket för tröghetskraft vektorn och tröghets-tangent matrisen. Balkelementet har sju frihetsgrader i varje nod och kubiska formfunktioner används för att interpolera lokala tvärgående förskjutningar och axiella rotationer. Fem exempel testas för att jämföra den nya formuleringen mot 3D-solidanalyser i Abaqus. Detta arbete presenteras i den fjärde bifogade tidskriftartikeln.

Résumé

L'objectif de ce travail de thèse est de développer des éléments poutres 2D et 3D dans le cadre co-rotationnel pour l'étude du comportement dynamique non-linéaire des structures à barres. Bien que la littérature contienne un certain nombre de références sur les poutres co-rotationnelles en statique, la formulation d'éléments poutres en dynamique dans un cadre co-rotationnel rigoureux reste un problème ouvert.

Dans le premier article, nous proposons une formulation co-rotationnelle robuste et efficace pour l'analyse dynamique non-linéaire des structures à barres planes. La principale nouveauté réside dans l'utilisation de fonctions d'interpolations cubiques à la fois pour la détermination des efforts internes mais aussi des termes d'inertie. En négligeant le carré du déplacement transversal dans le repère local, une expression analytique concise des termes dynamiques est obtenue. Cette nouvelle formulation est comparée aux deux formulations classiques utilisant la matrice masse de Timoshenko ou la matrice diagonale des masses concentrées. On peut souligner la bonne précision des résultats avec un nombre réduit d'éléments.

Le second article traite de la dynamique non-linéaire des poutres 3D pour lesquelles la paramétrisation des rotations finies et les méthodes d'intégration temporelle jouent un rôle important. Une étude bibliographique approfondie sur ces deux aspects, nous conduit à analyser et comparer quatre formulations dynamiques. Pour chacune d'entre elles, l'approche co-rotationnelle est utilisée pour développer les termes statiques, tandis que les termes dynamiques sont établis dans un contexte lagrangien total avec interpolations linéaires. Les calculs théoriques relatifs aux quatre formulations ainsi que la mise en œuvre pratique des différents algorithmes sont fournis en détail. La précision numérique des calculs et l'efficacité des formulations sont ensuite comparées. D'autre part, quatre prédicteurs et diverses possibilités de simplifier la matrice d'inertie tangente sont testés. La paramétrisation des rotations à l'aide de la partie vectorielle des quaternions est retenue et une simplification de la matrice tangente dynamique est proposée.

Dans le troisième article, une nouvelle formulation co-rotationnelle cohérente est développée pour l'analyse dynamique de poutre 3D (sans gauchissement). La nouveauté de cette formulation réside dans l'utilisation de la méthode co-rotationnelle pour établir l'expression non seulement du vecteur des efforts internes, de la matrice de rigidité tangente, mais aussi du vecteur des efforts d'inertie et la matrice d'inertie tangente. Ainsi, les fonctions d'interpolations cubiques sont adoptées pour tous les termes de l'équation de la dynamique. Une approximation des rotations locales est introduite permettant d'établir une expression concise du vecteur force d'inertie. Quatre exemples numériques sont considérés pour évaluer la performance de la nouvelle formulation par rapport à deux autres approches basées sur des interpolations linéaires.

Enfin, dans le quatrième article, l'élément de poutre co-rotationnel 3D précédent est étendu pour la dynamique non-linéaire des structures avec

une section transversale à paroi mince en introduisant les déformations de gauchissement et l'excentricité du centre de cisaillement. Cette cinématique conduit à des termes supplémentaires dans les expressions du vecteur force d'inertie et la matrice d'inertie tangente. L'élément a sept degrés de liberté au niveau de chaque nœud, et les fonctions de forme cubique sont utilisées pour interpoler les déplacements transversaux locaux et les rotations axiales. La performance de la formulation est évaluée à travers cinq exemples et des comparaisons avec des analyses éléments finis 3D réalisées avec le code Abaqus.

Mots-clés : méthode co-rotationnelle ; dynamique non-linéaire ; grands déplacements ; grandes rotations ; méthode d'intégration temporelles ; section transversale à paroi mince ; élément de poutre ;

Preface

The research presented in this PhD thesis was carried out both at the Division of Structural Engineering and Bridges, Department of Civil and Architectural Engineering, KTH Royal Institute of Technology, SWEDEN and at the Department of Civil Engineering, INSA de Rennes, FRANCE. The project was financed by Region Bretagne (FRANCE) and also by the Division of Structural Engineering and Bridges, KTH.

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Rennes, August 2013

Thanh-Nam Le

Publications

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Paper 1: Le, T.-N., Battini, J.-M. and Hjiaj, M., Efficient formulation for dynamics of corotational 2D beams.
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DOI: 10.1007/s00466-011-0585-6.

Paper 2: Le, T.-N., Battini, J.-M. and Hjiaj, M., Dynamics of 3D beam elements in a corotational context: a comparative study of established and new formulations.
Finite Elements in Analysis and Design, 61 : 97-111, 2012.
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Paper 3: Le, T.-N., Battini, J.-M. and Hjiaj, M., A consistent 3D corotational beam element for nonlinear dynamic analysis of flexible structures.
Submitted to *Computer Methods in Applied Mechanics and Engineering*, 2013.

Paper 4: Le, T.-N., Battini, J.-M. and Hjiaj, M., Corotational formulation for nonlinear dynamics of beams with arbitrary thin-walled open cross-sections.
Submitted to *Computers & Structures*, 2013.

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Chapter 1

Introduction

1.1 Background

Flexible beams are used in many applications, for instance large deployable space structures, aircrafts and wind turbines propellers, offshore platforms. These structures often undergo large displacements and finite rotations, but still with small deformations. The simulation of their nonlinear dynamic behavior is usually performed using beam finite elements. A considerable number of beam models related to this attractive topic can be found in the literature and most of these formulations are based on a classic total Lagrangian formulation or a corotational formulation.

The classic total Lagrangian formulation is based on the use of displacement and rotation variables with respect to a fixed inertial frame. Therefore, the kinetic energy takes a simple, quadratic form. This is the main reason why this method is frequently used to develop nonlinear beam models in dynamic analysis (see [52, 54] for 2D case, and [11, 30, 34, 35, 43, 55] for 3D case). However, as mentioned in [34], the nonlinear relations between global displacement and rotation variables and the strain measures introduce a complex definition of the strain energy, even in case of small strains. Furthermore, due to the fact that global displacement and rotation variables are considered, linear interpolations is the unique choice for two-noded beam formulations, which is not accurate, especially for flexible beams.

The corotational approach is also a total Lagrangian formulation, but the idea is to decompose the motion of the element into rigid body and pure deformational parts. During the rigid body motion, a local coordinate system, fixed to the element, moves and rotates with it. The deformational part is measured in this local system. Indeed, this framework has been adopted by several authors to develop efficient beam and shell elements for the nonlinear static and dynamic analysis of flexible structures [9, 14–17, 19, 24, 26, 33, 44, 46, 47, 56]. Moreover, numerous works based on the corotational approach have been carried out in the last decade by the Division of Structural Engineering

and Bridges (KTH-Stockholm) and by the Structural Engineering Research Group (INSA de Rennes). Several efficient 2D and 3D beam elements have been developed in order to model large displacements of structures in statics [1, 2, 5, 8, 49]. The structures can have solid or thin-walled cross-section.

The main interest of the corotational method is that with a proper choice of the length of the element, the local displacements are small relative to the local system. Therefore, different assumptions can be made to represent the local deformations, giving rise to different possibilities for the local element formulation.

For the geometrically and materially static nonlinear analysis of beam structures with corotational approach, several local formulations have been proposed by Battini and Pacoste [8], and Alsafadie et al. [1]. The results of a comparative study of 3D beam formulations, which can be found in [1], have shown that local beam elements based on cubic interpolations are more efficient and accurate than the ones which employ linear interpolations. However, in dynamics, one has to deal with the inertia terms which by nature are complicate to formulate. This is particularly true in the corotational formulation of Bernoulli-type beam elements. This difficulty has hampered the development of the corotational approach in nonlinear dynamics. To avoid the consistent derivation of the inertia terms, several routes have been considered.

For 2D dynamic analysis, quite a few authors [44, 46, 56] adopted the constant lumped mass matrix without any attempt to check its accuracy. Iura and Atluri [33] suggested to simply switch to a Timoshenko beam model where the mass matrix is constant and therefore the inertial terms are simple to evaluate. Behdinan et al. [9] proposed a 2D corotational dynamic formulation where cubic interpolations were used to describe the global displacements, which is not consistent with the idea of the corotational method as originally introduced by Nour-Omid and Rankin [45]. For 3D dynamic analysis, Crisfield et al. [14] suggested to use a constant Timoshenko mass matrix along with local cubic interpolations to derive the internal force vector and the corresponding tangent stiffness matrix. As pointed out by Crisfield et al. [14], this combination was not consistent but it provided reasonable results when the number of elements is large enough. Hsiao et al. [24] presented a corotational formulation for the nonlinear analysis of 3D beams. However the corotational framework adopted in [24] is different from the classic one [45]. In fact, to the author's knowledge, there is no general formulation for dynamic corotational beam elements in the literature. Hence, a consistent corotational beam formulation for 2D and 3D nonlinear dynamics is a very interesting topic for further research.

Finally, it must be noted that nonlinear 3D beam elements are not just a simple extension of 2D ones, mainly because of the complex nature of the finite rotations. More specifically, the finite rotations are non-commutative and non-additive. They cannot be treated in the same simple manner as translations and require special update and time stepping procedures. Several possibilities to parameterize the finite rotations can be found in [11, 30, 55]. Each parametrization has advantages and drawbacks, and the choice of an effective approach is still an issue.

1.2 Aims and scope

The main objective of this work is to extend the corotational beam elements in the previous works [1, 2, 5, 8, 49] by considering inertia effect and to investigate nonlinear dynamic behaviors of structures with large displacements. It should be noted that several versions of the corotational method have been proposed in the literature. The one employed in this work has been proposed by Rankin and Nour-Omid [45, 50], and then further developed by Battini and Pacoste [8].

For this purpose, the author first focuses on 2D beam elements. A general 2D corotational dynamic beam element is developed. In order to capture correctly all inertia effects, the local cubic interpolations are adopted not only to obtain the elastic terms but also to derive the inertia terms. Several numerical examples are then implemented to compare the accuracy of the new formulation against two classic formulations using linear Timoshenko mass matrix and linear lumped mass matrix. This work is presented in the first paper [37].

Before extending the previous 2D dynamic beam element to 3D, the parametrization of the finite rotations and their update procedures for nonlinear 3D beam elements are carefully investigated. For that, four dynamic beam formulations derived in a total Lagrangian context together with corotational internal force vectors and tangent stiffness matrices are compared. They are based on three different parameterizations of rotations. The first three formulations are taken from the literature. The last one is new and uses three of the four Euler parameters (quaternion) as rotational variables. For all these approaches, theoretical derivations as well as practical implementations are given in detail. The similarities and differences between them are pointed out. Six numerical examples are then implemented in order to compare these four formulations in terms of numerical accuracy and computational efficiency. Regarding efficiency, several predictors and various possibilities to simplify the tangent inertia matrix are tested. This work is presented in the second paper [38].

The next part of this PhD work concerns 3D corotational dynamic beam formulations. First, a 3D dynamic beam element is developed. The same idea as in case of 2D is used: to derive both internal and inertia terms, the corotational method and local cubic interpolations are adopted. This approach ensures the consistency of the formulation. A proper approximation for the local rotations is adopted in the derivations of the inertia terms. To enhance the efficiency of the iterative procedure, the less significant term in the tangent dynamic matrix is ignored. Four numerical examples are then analysed with the objective to compare the performances of the new formulation against two other approaches. The first one is similar to the one presented above, but linear local interpolations, instead of cubic ones, are used to derive only the dynamic terms. The purpose is to evaluate the influence of the choice of the local interpolations on the dynamic terms. The second approach is the classic total Lagrangian formulation proposed by Simo and Vu-Quoc [53–55]. This work is presented in the third paper [39].

Finally, the previous 3D beam element is extended to model beams with arbitrary thin-

walled open cross-sections. For this purpose, warping deformations and eccentricity of shear center are taken into account. In order to introduce the warping deformations, the kinematic description proposed by Gruttmann et al. [22] is adopted. Consequently, the beam element has seven degrees of freedom at each node. Regarding the static deformational terms, i.e. the internal force vector and tangent stiffness matrix, the corotational beam element developed by Battini and Pacoste [8] is adopted. However, in order to introduce the bending shear deformations, the cubic Hermitian functions are modified as suggested in the Interdependent Interpolation Element (IIE) formulation [51]. Five numerical examples are analysed to evaluate the accuracy of the formulation against Abaqus 3D-solid elements. This work is presented in the fourth paper [40].

1.3 Outline of thesis

This thesis is based on the work and results given in the four appended journal papers. To highlight their contributions and to put them in the context of an overall project, relevant aspects of each paper are summarized in the four following chapters. Moreover, to make this thesis self-contained, additional knowledge is introduced. The organisation of the thesis is as follows

Chapter 2 presents main features of the 2D beam formulation, which was developed in the first paper [37].

Chapter 3 summarizes important aspects which were dealt in the second paper [38].

Chapter 4 presents in detail 3D corotational beam kinematics, derivations of internal force vectors and tangent stiffness matrices for different parameterizations of finite rotations and several local beam elements. This presentation is taken entirely from the work of Battini [5, 6]. Then, the dynamic formulation developed in the third paper [39] is briefly introduced.

Chapter 5 gives a presentation of a corotational formulation for beams with arbitrary thin-walled cross-sections. This formulation was developed in the fourth paper [40].

Chapter 2

2D corotational beam element

The corotational approach is a well known method to derive efficient nonlinear beam finite elements [9, 14, 16, 17, 19, 24, 26, 33, 44, 46, 56]. The main idea is to decompose the motion of the element into rigid body and pure deformational parts. During the rigid body motion, a local coordinates system, fixed to the element, moves and rotates with it. The deformational part is measured in this local system. The main interest of the approach is that different assumptions can be made to represent the local deformations, giving different possibilities for the local element formulation.

Regarding the dynamic formulation of 2D corotational beams, several options are available. If linear interpolations are used for the local formulation, e.g. by taking the classical linear Timoshenko element, then inertia corotational terms are easily derived and the classical linear and constant Timoshenko mass matrix is obtained. However, linear interpolations assume that the transverse displacements are equal to zero along the element, which is not accurate, especially for flexible beams. If cubic interpolations are used for the local formulation, e.g. by taking the classic linear Bernoulli element, then the derivation of the inertia terms becomes very complicated. In [14], Crisfield et al. suggested that this derivation is impossible due to its complexity. Therefore, they used the constant Timoshenko mass matrix although they adopted local cubic interpolations to derive the elastic force vector and the tangent stiffness matrix. The same approach was adopted in [33]. In [44, 46, 56], the authors used a constant lumped mass matrix without any attempt to check its accuracy. In [9], Behdinan et al. proposed a corotational dynamic formulation. But the cubic shape functions were used to describe the global displacements, which is not consistent with the idea of the corotational method.

In the paper [37], the authors proposed a new corotational formulation. The novelty is that local cubic interpolations were used not only to obtain the elastic terms but also to derive the inertia terms. The efficiency of the formulation, compared with the two other corotational approach based on Timoshenko and lumped mass matrix, was shown through four numerical examples. The purpose of this chapter is to summarize relevant aspects of this formulation.

2.1 Corotational framework

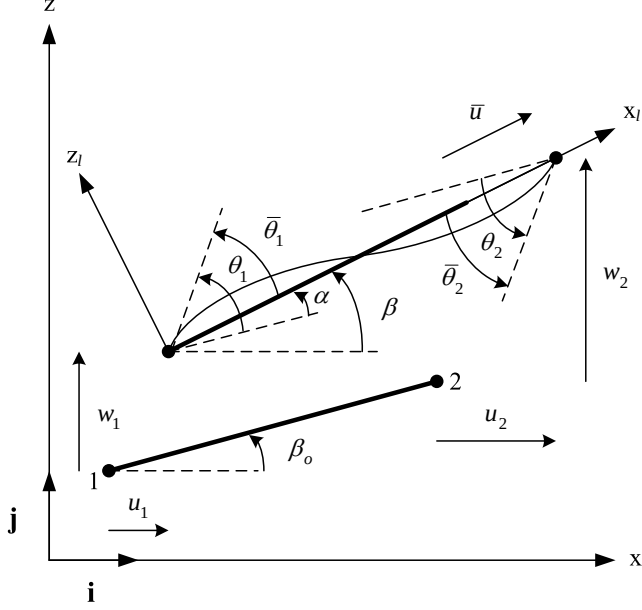


Fig. 2.1: Beam kinematics 1.

The corotational beam kinematic for a two-noded straight element is presented in Fig. 2.1. The coordinates for the nodes 1 and 2 in the global coordinate system (x, z) are (x_1, z_1) and (x_2, z_2) . The vector of global displacements is defined by

$$\mathbf{q} = [u_1 \quad w_1 \quad \theta_1 \quad u_2 \quad w_2 \quad \theta_2]^T, \quad (2.1)$$

while the vector of local displacements is defined by

$$\bar{\mathbf{q}} = [\bar{u} \quad \bar{\theta}_1 \quad \bar{\theta}_2]^T. \quad (2.2)$$

The components of $\bar{\mathbf{q}}$ can be computed according to

$$\bar{u} = l_n - l_o, \quad (2.3)$$

$$\bar{\theta}_1 = \theta_1 - \alpha = \theta_1 - \beta + \beta_o, \quad (2.4)$$

$$\bar{\theta}_2 = \theta_2 - \alpha = \theta_2 - \beta + \beta_o. \quad (2.5)$$

In Eq. (2.3), l_o and l_n denote the initial and current lengths of the element

$$l_o = [(x_2 - x_1)^2 + (z_2 - z_1)^2]^{1/2}, \quad (2.6)$$

$$l_n = [(x_2 + u_2 - x_1 - u_1)^2 + (z_2 + w_2 - z_1 - w_1)^2]^{1/2}. \quad (2.7)$$

The current angle of the local system with respect to the global system is denoted as β and is given by

$$c = \cos \beta = \frac{1}{l_n}(x_2 + u_2 - x_1 - u_1) , \quad (2.8)$$

$$s = \sin \beta = \frac{1}{l_n}(z_2 + w_2 - z_1 - w_1) . \quad (2.9)$$

At the initial configuration, $\beta = \beta_o$ and

$$c_o = \cos \beta_o = \frac{1}{l_n}(x_2 - x_1) , \quad (2.10)$$

$$s_o = \sin \beta_o = \frac{1}{l_n}(z_2 - z_1) . \quad (2.11)$$

The rigid rotation α is computed as

$$\sin \alpha = c_o s - s_o c , \quad (2.12)$$

$$\cos \alpha = c_o c - s_o s . \quad (2.13)$$

The local rotations $\bar{\theta}_i$ ($i = 1, 2$) are calculated using

$$\sin \bar{\theta}_i = \sin \theta_i \cos \alpha - \cos \theta_i \sin \alpha , \quad (2.14)$$

$$\cos \bar{\theta}_i = \cos \theta_i \cos \alpha + \sin \theta_i \sin \alpha . \quad (2.15)$$

Then, $\bar{\theta}_i$ is determined by

$$\begin{aligned} \bar{\theta}_i &= \sin^{-1}(\sin \bar{\theta}_i) && \text{if } \sin \bar{\theta}_i \geq 0 \text{ and } \cos \bar{\theta}_i \geq 0 , \\ \bar{\theta}_i &= \cos^{-1}(\cos \bar{\theta}_i) && \text{if } \sin \bar{\theta}_i \geq 0 \text{ and } \cos \bar{\theta}_i < 0 , \\ \bar{\theta}_i &= \sin^{-1}(\sin \bar{\theta}_i) && \text{if } \sin \bar{\theta}_i < 0 \text{ and } \cos \bar{\theta}_i \geq 0 , \\ \bar{\theta}_i &= -\cos^{-1}(\cos \bar{\theta}_i) && \text{if } \sin \bar{\theta}_i < 0 \text{ and } \cos \bar{\theta}_i < 0 . \end{aligned} \quad (2.16)$$

It should be noted that this procedure to derive $\bar{\theta}_i$ differs from the one in [13]. Here, the angles of rotations are not limited by π as in [13].

The connection between the variations of the local and global displacements is obtained as

$$\delta \bar{\mathbf{q}} = \mathbf{B} \delta \mathbf{q} . \quad (2.17)$$

By taking the differentiation of the expressions (2.3) to (2.5), the transformation matrix $\mathbf{B}$ is obtained as

$$\mathbf{B} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{bmatrix} = \begin{bmatrix} -c & -s & 0 & c & s & 0 \\ -s/l_n & c/l_n & 1 & s/l_n & -c/l_n & 0 \\ -s/l_n & c/l_n & 0 & s/l_n & -c/l_n & 1 \end{bmatrix} . \quad (2.18)$$

2.2 Local beam kinematic description

The Interdependent Interpolation Element (IIE), proposed in [51], is adopted for the local beam kinematic description. The development of this beam element is based on the exact solution of the homogeneous form of the equilibrium equations for a Timoshenko beam. Consequently, the IIE element not only retains the accuracy inherent to the cubic interpolation, but also includes the bending shear deformation. The shape functions of the IIE element are given by

$$\varphi_1 = \mu x \left[6\Omega \left(1 - \frac{x}{l_o} \right) + \left(1 - \frac{x}{l_o} \right)^2 \right], \quad (2.19)$$

$$\varphi_2 = \mu x \left[6\Omega \left(\frac{x}{l_o} - 1 \right) - \frac{x}{l_o} + \frac{x^2}{l_o^2} \right], \quad (2.20)$$

$$\varphi_3 = \mu \left(1 + 12\Omega - \frac{12\Omega x}{l_o} - \frac{4x}{l_o} + \frac{3x^2}{l_o^2} \right), \quad (2.21)$$

$$\varphi_4 = \mu \left(\frac{12\Omega x}{l_o} - \frac{2x}{l_o} + \frac{3x^2}{l_o^2} \right), \quad (2.22)$$

where

$$\Omega = \frac{EI}{GA K_s l_o^2}, \quad \mu = \frac{1}{1 + 12\Omega}, \quad (2.23)$$

A, I : Section's area and second moment of area ,

K_s : Shear correction coefficient .

With $\Omega = 0$, the Hermitian shape functions of the classical Bernoulli elements are recovered.

Using this beam element, the axial displacement u , the transverse displacement w and the local rotation ϑ are calculated by

$$u = \frac{x}{l_o} \bar{u}, \quad (2.24)$$

$$w = \varphi_1 \bar{\theta}_1 + \varphi_2 \bar{\theta}_2, \quad (2.25)$$

$$\vartheta = \varphi_3 \bar{\theta}_1 + \varphi_4 \bar{\theta}_2. \quad (2.26)$$

2.3 Elastic force vector and tangent stiffness matrix

By equating the virtual work in the local and global systems, the relation between the local elastic force vector $\mathbf{f}_l$ and the global one $\mathbf{f}_g$ is obtained as

$$V = \delta \mathbf{q}^T \mathbf{f}_g = \delta \bar{\mathbf{q}}^T \mathbf{f}_l = \delta \mathbf{q}^T \mathbf{B}^T \mathbf{f}_l. \quad (2.27)$$

Eq. (2.27) must apply for any arbitrary $\delta \mathbf{q}$. Hence the global elastic force vector $\mathbf{f}_g$ is given by

$$\mathbf{f}_g = \mathbf{B}^T \mathbf{f}_l, \quad \text{with} \quad \mathbf{f}_l = \begin{bmatrix} N & M_1 & M_2 \end{bmatrix}^T. \quad (2.28)$$

The global tangent stiffness matrix is defined by

$$\delta \mathbf{f}_g = \mathbf{K}_g \delta \mathbf{q}. \quad (2.29)$$

By taking the differentiation of Eq. (2.28), the global stiffness matrix is obtained as

$$\mathbf{K}_g = \mathbf{B}^T \mathbf{K}_l \mathbf{B} + \frac{\mathbf{z} \mathbf{z}^T}{l_n} N + \frac{1}{l_n^2} (\mathbf{r} \mathbf{z}^T + \mathbf{z} \mathbf{r}^T) (M_1 + M_2), \quad (2.30)$$

where

$$\mathbf{r} = \begin{bmatrix} -c & -s & 0 & c & s & 0 \end{bmatrix}^T, \quad (2.31)$$

$$\mathbf{z} = \begin{bmatrix} s & -c & 0 & -s & c & 0 \end{bmatrix}^T. \quad (2.32)$$

The local elastic force vector $\mathbf{f}_l$ and local tangent stiffness matrix $\mathbf{K}_l$, which is defined by $\delta \mathbf{f}_l = \mathbf{K}_l \delta \bar{\mathbf{q}}$, depend on the definition of the local formulation.

The shape functions of the IIE were used together with a shallow arch beam theory. The shallow arch longitudinal and shear strains are given by

$$\varepsilon = \frac{1}{l_o} \int_{l_o} \left[\frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2 \right] dx - \frac{\partial^2 w}{\partial x^2} z, \quad (2.33)$$

$$\gamma = \frac{\partial w}{\partial x} + \vartheta. \quad (2.34)$$

The expressions of $\mathbf{f}_l$, $\mathbf{K}_l$ were then derived using Maple. The code was given in [37].

2.4 Inertia force vector and tangent dynamic matrix

The kinetic energy K of an element is given as

$$K = \frac{1}{2} \rho \left\{ \int_{l_o} A (\dot{u}_G^2 + \dot{w}_G^2) dl + \int_{l_o} I \dot{\theta}^2 dl \right\}, \quad (2.35)$$

where

- ρ : Mass per unit volume,
- u_G, w_G : Global displacements of the cross-section centroid,
- θ : Global rotation of the section.

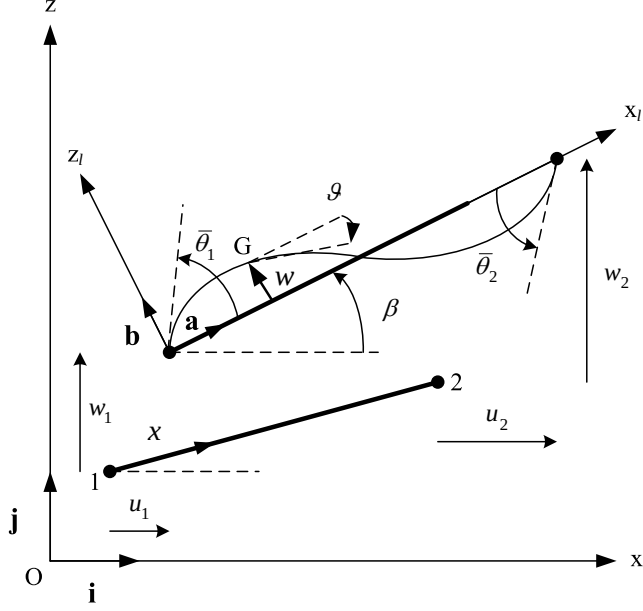


Fig. 2.2: Beam kinematics 2.

The global position of the cross-section centroid is given by (see Fig. 2.2)

$$\mathbf{OG} = (x_1 + u_1)\mathbf{i} + (z_1 + w_1)\mathbf{j} + \frac{l_n}{l_o}x\mathbf{a} + w\mathbf{b}, \quad (2.36)$$

The velocities of the centroid are then calculated by taking time derivative of the previous expression

$$\dot{u}_G = \dot{u}_1 + \frac{x}{l_o}(\dot{u}_2 - \dot{u}_1) - \dot{w}\sin\beta - w\dot{\beta}\cos\beta, \quad (2.37)$$

$$\dot{w}_G = \dot{w}_1 + \frac{x}{l_o}(\dot{w}_2 - \dot{w}_1) + \dot{w}\cos\beta - w\dot{\beta}\sin\beta, \quad (2.38)$$

The global rotation of the cross-section is given by

$$\dot{\theta} = \dot{\vartheta} + \dot{\alpha} = \dot{\vartheta} + \dot{\beta}. \quad (2.39)$$

The quantities $w, \dot{w}, \dot{\vartheta}$ are interpolated from the nodal quantities using Eqs. (2.25) and (2.26). However, Ω is taken to 0 as suggested in [51]. Extensive numerical studies performed have shown that this simplification does not modify the numerical results.

The exact expression of the kinetic energy K is then obtained by substituting Eqs. (2.37)-(2.39) into Eq. (2.35).

Moreover, the kinetic energy can be written as

$$K = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{T}^T \mathbf{M}_l \mathbf{T} \dot{\mathbf{q}}, \quad (2.40)$$

where $\mathbf{T}$ is rotation matrix. Hence, the expression of the local matrix $\mathbf{M}_l$ can be derived. At this point, two simplifications are introduced in the expression of the local mass matrix. The local displacement w is assumed small and therefore the terms containing w^2 are neglected. Furthermore, the approximation $l_n = l_o$ is considered (small axial deformation assumption). With these simplifications, the local mass matrix is only function of $\bar{\theta}_1$ and $\bar{\theta}_2$.

The inertia force vector is calculated from the kinetic energy by using the Lagrange's equation of motion

$$\mathbf{f}_K = \frac{d}{dt} \left[\frac{\partial K}{\partial \dot{\mathbf{q}}} \right] - \left[\frac{\partial K}{\partial \mathbf{q}} \right]. \quad (2.41)$$

With the expression of the kinetic energy, $\mathbf{f}_K$ is obtained as

$$\begin{aligned} \mathbf{f}_K = \mathbf{M} \ddot{\mathbf{q}} + & \left\{ \mathbf{M}_\beta \left(\frac{\mathbf{z}^T}{l_n} \dot{\mathbf{q}} \right) + \mathbf{M}_{\bar{\theta}_1} (\mathbf{b}_2^T \dot{\mathbf{q}}) + \mathbf{M}_{\bar{\theta}_2} (\mathbf{b}_3^T \dot{\mathbf{q}}) \right\} \dot{\mathbf{q}} \\ & - \left(\frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}_\beta \dot{\mathbf{q}} \right) \frac{\mathbf{z}}{l_n} - \left(\frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}_{\bar{\theta}_1} \dot{\mathbf{q}} \right) \mathbf{b}_2 - \left(\frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}_{\bar{\theta}_2} \dot{\mathbf{q}} \right) \mathbf{b}_3, \end{aligned} \quad (2.42)$$

where $\mathbf{M}_\beta = \frac{\partial \mathbf{M}}{\partial \beta}$, $\mathbf{M}_{\bar{\theta}_1} = \frac{\partial \mathbf{M}}{\partial \bar{\theta}_1}$, $\mathbf{M}_{\bar{\theta}_2} = \frac{\partial \mathbf{M}}{\partial \bar{\theta}_2}$.

From the expression of the inertia force vector given in Eq. (2.42), the following differentiations are calculated

$$\mathbf{C}_K = \frac{\partial \mathbf{f}_K}{\partial \dot{\mathbf{q}}}, \quad (2.43)$$

$$\mathbf{K}_K = \frac{\partial \mathbf{f}_K}{\partial \mathbf{q}}. \quad (2.44)$$

For the expressions of $\mathbf{C}_K$ and $\mathbf{K}_K$, the interested reader is referred to [37].

2.5 Numerical example - Cantilever beam

The example, described in Fig. 2.3, is a cantilever beam of length $L = 10$ m with uniform cross-section and subjected to a sinusoidal tip force $P = P_o \sin(\omega t)$ at the free end. The amplitude of the load P_o is taken equal to 10 MN and its frequency ω is 50 rad/s. The cross-section depth and width are $a = 0.25$ m and $e = 0.5$ m, respectively. The elastic modulus of the beam E is 210 GPa and the mass per unit volume is $\rho = 7850$ kg/m³.

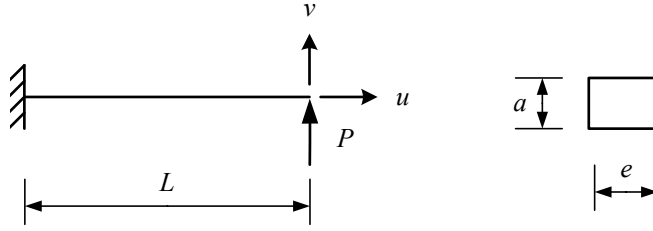


Fig. 2.3: Cantilever beam : geometrical data.

The nonlinear dynamic behavior of the beam was analysed using three different beam formulations: the new one, and the two formulations usually found in the literature, i.e. the lumped mass matrix and the Timoshenko mass matrix. For all dynamic formulations, the elastic force vector and tangent stiffness matrix were derived using the IIE shape functions in order to account for shear deformability. The three dynamic formulations were compared with a reference solution. This solution was obtained with a large number of elements and was identical for the three considered dynamic formulations. The reference solution was also checked with Abaqus (total Lagrangian formulation) and the same results were obtained.

In this example, the reference solution was obtained with 48 elements. The results given by three formulations considering only 3 elements, with the time step size $\Delta t = 10^{-4}$ s, were shown in Figs. 2.4 and 2.5. For the presentation of the results, the following colors were used in all figures:

-----	Reference solution	—	Lumped mass matrix
- - - - -	Timoshenko mass matrix	—	New formulation

It can be observed that the proposed formulation gave results that were in very agreement with the reference solution with only three elements. However, the results obtained with the lumped and Timoshenko approaches did not agree well with the reference solution over the whole time domain.

This example shows that the new formulation, based on local cubic interpolations, is more efficient than two other formulations, which are based on local linear interpolations (Timoshenko mass matrix) and lumped mass matrix. This advantage is due to a better representation of the local displacements in the inertia terms.

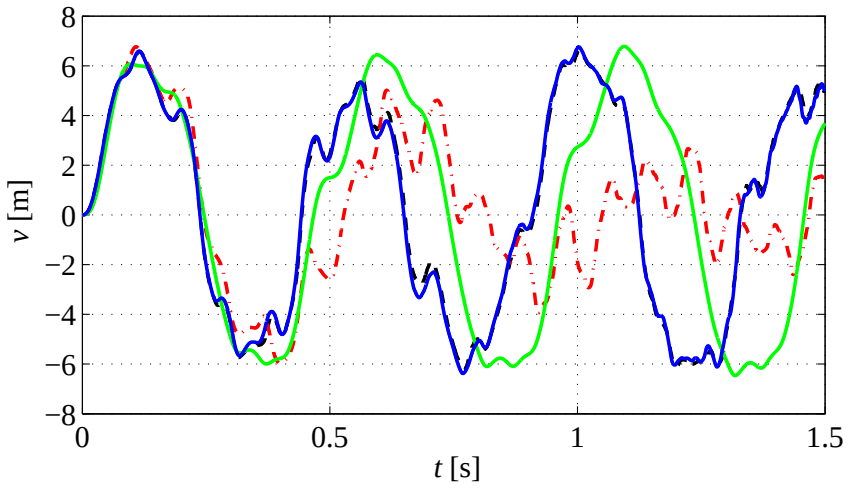


Fig. 2.4: Cantilever beam - Vertical displacement history.

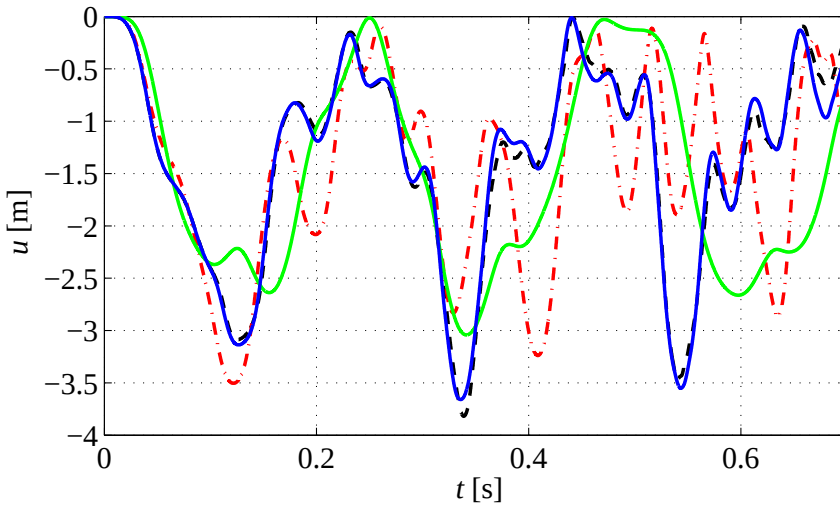


Fig. 2.5: Cantilever beam - Horizontal displacement history.

Chapter 3

Finite rotations in dynamics

The extension of a 2D beam formulation to 3D is not an easy task. The difficulty is mainly due to the complex nature of the finite rotations. More specifically, the finite rotations are non-commutative and non-additive, thus they cannot be treated in a simple manner as the translational displacements. As a consequence, the standard Newmark time stepping method cannot be directly applied to the finite rotations. This method must be reformulated according to the parametrization of the finite rotations.

Several aspects of the finite rotations in dynamics were investigated in [38]. The first one concerned the parametrization of the finite rotations. In the computational mechanics, many possibilities to parameterize finite rotations can be found [3, 11, 13, 21, 36]. Rotation tensor, rotational vector and Euler parameters were studied. These parameterizations are popular in the finite element method, however, each of them has certain deficiencies. The rotation tensor, with nine components to handle, requires special update procedure. In contrast, only three parameters are needed for the rotational vector, and the Euler parameters. Furthermore, the rotations become additive and can be updated in the same manner as the translations. However, the amplitude of the finite rotations represented by these parameterizations are limited by 2π for the rotational vector, and by π for the Euler parameters. To avoid this inconvenience, Cardona and Geradin [11], Ibrahimbegović [28] and Battini [7] introduced the notion of the incremental rotational vector and the incremental Euler parameters. Then, additive updates still apply but this time only within each increment. The amplitudes of the rotations are just limited within each increment, which is not a problem.

The second aspect was related to the Newmark time stepping methods for the finite rotations and the efficiency of the iterative procedure. Two Newmark algorithms have been used in the literature. In the first one, proposed by Simo and Vu-Quoc [55], the Newmark equations were written using the material incremental rotational vector, the material angular velocity and the material acceleration. This approach was adopted in [14, 32, 34, 35]. Ibrahimbegović and Mikdad [30] reformulated this method using spatial forms. In the second approach, introduced by Cardona and Geradin [11], the standard Newmark algorithm

was applied to the incremental rotational vector and its time derivatives. Hence, the update procedure of the rotational quantities took a similar form as the displacements. In the works of Forsell [18] and Mäkinen [43], this approach was adopted using the total rotational vectors. Regarding the efficiency of the iterative procedure, several predictors and possibilities to simplify the tangent inertia matrix were also tested.

In order to compare these parameterizations and the Newmark methods for the finite rotations, four nonlinear dynamic formulations were investigated in [38]. For all of these formulations, theoretical derivations and practical implementations were given in detail. The similarities and differences between them were also pointed out. Moreover, to assess the four formulations in terms of numerical accuracy and computational efficiency, six nonlinear dynamic examples were analyzed.

The objective of this chapter is to present a short summary of these aspects and one example is reproduced. The organisation of the chapter is as follows. Section 3.1 presents three parameterizations of the finite rotations. The Newmark time stepping methods are introduced in Section 3.2. The four dynamic formulations are summarized in Section 3.3. Finally, one numerical example is investigated in Section 3.4.

3.1 Finite rotations

Parametrization of the finite rotations

The coordinate of a fixed vector $\mathbf{x}_o$ that is rotated into the position $\mathbf{x}$ (see Fig. 3.1) is given by the relation

$$\mathbf{x} = \mathbf{R} \mathbf{x}_o. \quad (3.1)$$

The rotation matrix $\mathbf{R}$ is an element of the $SO(3)$ group. Its coordinate representation is

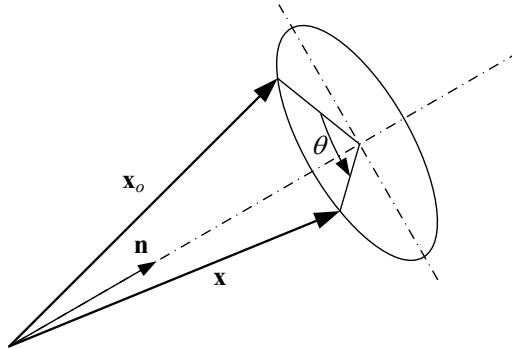


Fig. 3.1: Finite rotation of a vector.

a 3x3 orthogonal matrix involving nine components. However, due to its orthonormality, the rotation matrix $\mathbf{R}$ can be parameterized using only three independent parameters. One possibility is to use the rotational vector defined by

$$\boldsymbol{\theta} = \theta \mathbf{n}, \quad (3.2)$$

where $\mathbf{n}$ is a unit vector defining the axis of the rotation and $\theta = (\boldsymbol{\theta}^T \boldsymbol{\theta})^{1/2}$ is the angle of the rotation.

The relation between the rotation matrix and the rotational vector is given by the Rodrigues' formula

$$\mathbf{R} = \mathbf{I} + \frac{\sin \theta}{\theta} \tilde{\boldsymbol{\theta}} + \frac{1 - \cos \theta}{\theta^2} \tilde{\boldsymbol{\theta}} \tilde{\boldsymbol{\theta}}, \quad (3.3)$$

where $\tilde{\boldsymbol{\theta}}$ is the skew matrix associated to the vector $\boldsymbol{\theta}$.

If the trigonometric functions in the above equation are expanded in Taylor series, $\mathbf{R}$ can be written as

$$\mathbf{R} = \mathbf{I} + \tilde{\boldsymbol{\theta}} + \frac{1}{2} \tilde{\boldsymbol{\theta}}^2 + \dots = \exp(\tilde{\boldsymbol{\theta}}). \quad (3.4)$$

Another possibility to parameterize the rotation matrix is to use the Euler parameters (quaternion), which by introducing the scaling factor 2 gives

$$\mathbf{q} = 2 \sin \frac{\theta}{2} \mathbf{n}. \quad (3.5)$$

In terms of $\mathbf{q}$, the rotational tensor $\mathbf{R}$ is given by

$$\mathbf{R}(\mathbf{q}) = (2q_o^2 - 1)\mathbf{I} + \frac{1}{2}\mathbf{q}\mathbf{q}^T + q_o\tilde{\mathbf{q}}, \quad (3.6)$$

with $q_o = (1 - \mathbf{q}^T \mathbf{q}/4)^{1/2}$.

It can be observed that this parametrization requires only three parameters $\mathbf{q}$ (see Eq. (3.5)) instead of four($\mathbf{q}, q_o$) as usually found in the literature.

Variation of the rotation parameters

The admissible variation $\delta \mathbf{R}$ of the rotation matrix $\mathbf{R}$ is calculated by

$$\begin{aligned} \delta \mathbf{R} &= \frac{d}{d\varepsilon} [\mathbf{R}_\varepsilon]_{\varepsilon=0} = \frac{d}{d\varepsilon} \left(\exp(\varepsilon \tilde{\boldsymbol{\omega}}) \mathbf{R} \right)_{\varepsilon=0} = \tilde{\boldsymbol{\omega}} \mathbf{R} \\ &= \frac{d}{d\varepsilon} \left(\mathbf{R} \exp(\varepsilon \tilde{\boldsymbol{\omega}}) \right)_{\varepsilon=0} = \mathbf{R} \tilde{\boldsymbol{\omega}}. \end{aligned} \quad (3.7)$$

Physically, $\delta \mathbf{w}$ and $\delta \boldsymbol{\omega}$ represent infinitesimal spatial and material rotations superposed onto the rotation $\mathbf{R}$. $\delta \mathbf{w}$, which is also denoted as spatial spin variables, is related to small variation of the rotational vector through

$$\delta \mathbf{w} = \mathbf{T}_s(\boldsymbol{\theta}) \delta \boldsymbol{\theta}, \quad (3.8)$$

with

$$\mathbf{T}_s(\boldsymbol{\theta}) = \mathbf{I} + \frac{1 - \cos \theta}{\theta^2} \boldsymbol{\theta} \boldsymbol{\theta}^T + \frac{\theta - \sin \theta}{\theta^3} \boldsymbol{\theta} \boldsymbol{\theta}^T. \quad (3.9)$$

In connection with the operator $\mathbf{T}_s$, it should also be noted that

$$\det(\mathbf{T}_s) = \frac{2(1 - \cos \theta)}{\theta^2}, \quad (3.10)$$

which shows that the corresponding mapping ceases to be a bijection for $\theta = 2k\pi$ ($k = 1, 2, \dots$) [28, 48].

The inverse relation of Eq. (3.8) is defined as

$$\delta \boldsymbol{\theta} = \mathbf{T}_s^{-1}(\boldsymbol{\theta}) \delta \mathbf{w}, \quad (3.11)$$

with

$$\mathbf{T}_s^{-1}(\boldsymbol{\theta}) = \frac{(\theta/2)}{\tan(\theta/2)} \mathbf{I} + \left(1 - \frac{(\theta/2)}{\tan(\theta/2)}\right) \frac{\boldsymbol{\theta} \boldsymbol{\theta}^T}{\theta^2} - \frac{1}{2} \boldsymbol{\theta} \boldsymbol{\theta}^T. \quad (3.12)$$

The relation between the spatial spin variables and a small variation of the Euler parameters is given by

$$\delta \mathbf{w} = \mathbf{T}_q(\mathbf{q}) \delta \mathbf{q}, \quad (3.13)$$

with

$$\mathbf{T}_q(\mathbf{q}) = q_o \mathbf{I} + \frac{\mathbf{q} \mathbf{q}^T}{4 q_o} + \frac{\tilde{\mathbf{q}}}{2}. \quad (3.14)$$

It should be noted that the operator $\mathbf{T}_q(\mathbf{q})$ is undefined for $q_o = 0$, which corresponds to $\theta = k\pi$.

Similar relations connecting the material spin variables $\delta \boldsymbol{\omega}$ and $\delta \boldsymbol{\theta}$, $\delta \mathbf{q}$ can also be obtained by noting that $\delta \mathbf{w} = \mathbf{R} \delta \boldsymbol{\omega}$. This issue, however will not be dealt with in this thesis.

Update procedures of finite rotations

Due to the fact that the finite rotations are not elements of a linear space, the two successive rotations are not commutative. Therefore, the update procedure of the finite rotations

after each Newton-Raphson iteration depends on the choice of the parametrization of finite rotations, and needs to be carefully treated.

If the spatial spin variables are used to parameterize finite rotations, the update of the rotation matrix at i^{th} iteration of step $(n+1)$ is performed according to

$$\mathbf{R}_{n+1}^i = \exp(\widetilde{\Delta \mathbf{w}}) \mathbf{R}_{n+1}^{i-1}, \quad (3.15)$$

where $\Delta \mathbf{w}$ are the iterative spatial spin variables.

If the rotational vector or the Euler parameters are used as parameters (see [6, 47]), then the rotations become additive and are simply updated at each iteration using

$$\boldsymbol{\theta}^i = \boldsymbol{\theta}^{i-1} + \Delta \boldsymbol{\theta}, \quad (3.16)$$

$$\mathbf{q}^i = \mathbf{q}^{i-1} + \Delta \mathbf{q}. \quad (3.17)$$

However, the relations in Eqs. (3.8) and (3.13) cease to be bijections respectively for $\theta = 2\pi$ and for $\theta = \pi$. Consequently, the angle of rotation is limited to 2π with the parametrization using the rotational vector and π in case of the Euler parameters. In many dynamic analysis, angles of rotations can become larger than these limitations.

In order to overcome this inconvenience, Cardona and Geradin [11], Ibrahimbegović [28] and Battini [7] proposed to perform Eqs. (3.8) and (3.13) only within an increment, and introduced the concept of incremental rotational vector and incremental Euler parameters. Then, the update procedure at the step $(n+1)$ is performed in the following way: at the beginning of the step, the incremental rotational vector and the incremental Euler parameters are set equal to zero ($\boldsymbol{\theta}_{n+1}^0 = \mathbf{q}_{n+1}^0 = \mathbf{0}_{[3 \times 1]}$). At iteration i , they are updated using

$$\boldsymbol{\theta}_{n+1}^i = \boldsymbol{\theta}_{n+1}^{i-1} + \Delta \boldsymbol{\theta}, \quad (3.18)$$

$$\mathbf{q}_{n+1}^i = \mathbf{q}_{n+1}^{i-1} + \Delta \mathbf{q}, \quad (3.19)$$

and the rotation matrix is updated using

$$\mathbf{R}_{n+1}^i = \exp(\widetilde{\boldsymbol{\theta}_{n+1}^i}) \mathbf{R}_n, \quad (3.20)$$

$$\mathbf{R}_{n+1}^i = \mathbf{R}(\mathbf{q}_{n+1}^i) \mathbf{R}_n, \quad (3.21)$$

where $\boldsymbol{\theta}_{n+1}^i, \mathbf{q}_{n+1}^i$ are the spatial incremental rotational vector and the spatial incremental Euler parameters, respectively.

Hence, additive updates still apply within each increment. The amplitude of the rotations are just limited in each increment, which is not a problem.

Angular velocities and accelerations

In order to simulate dynamic behavior of beam structures, angular velocities and accelerations must be calculated. Considering Eq. (3.7), the angular velocity can be expressed in

spatial form as

$$\tilde{\mathbf{w}} = \dot{\mathbf{R}} \mathbf{R}^T, \quad (3.22)$$

or in material form as

$$\tilde{\mathbf{\omega}} = \mathbf{R}^T \dot{\mathbf{R}}. \quad (3.23)$$

Due to the fact that the rotation matrix is an orthogonal matrix, the mutual relations between the corresponding axial vectors can be written as

$$\dot{\mathbf{w}} = \mathbf{R} \dot{\mathbf{\omega}}, \quad (3.24)$$

$$\dot{\mathbf{\omega}} = \mathbf{R}^T \dot{\mathbf{w}}. \quad (3.25)$$

By taking the time derivatives of Eqs. (3.22) and (3.23), the angular accelerations in spatial and material form are given by

$$\tilde{\dot{\mathbf{w}}} = \ddot{\mathbf{R}} \mathbf{R}^T + \dot{\mathbf{R}} \dot{\mathbf{R}}^T, \quad (3.26)$$

$$\tilde{\dot{\mathbf{\omega}}} = \mathbf{R}^T \ddot{\mathbf{R}} + \dot{\mathbf{R}}^T \dot{\mathbf{R}}. \quad (3.27)$$

The relations between the two forms of angular acceleration are given by

$$\dot{\mathbf{w}} = \mathbf{R} \dot{\mathbf{\omega}}, \quad (3.28)$$

$$\dot{\mathbf{\omega}} = \mathbf{R}^T \dot{\mathbf{w}}. \quad (3.29)$$

The spatial angular velocity can be directly calculated from the rotational vector and from the Euler parameters by

$$\dot{\mathbf{w}} = \mathbf{T}_s(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} = \mathbf{T}_q(\mathbf{q}) \dot{\mathbf{q}}. \quad (3.30)$$

By taking the time derivative of the previous equation, the spatial angular acceleration is calculated by

$$\ddot{\mathbf{w}} = \dot{\mathbf{T}}_s(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} + \mathbf{T}_s(\boldsymbol{\theta}) \ddot{\boldsymbol{\theta}} = \dot{\mathbf{T}}_q(\mathbf{q}) \dot{\mathbf{q}} + \mathbf{T}_q(\mathbf{q}) \ddot{\mathbf{q}}. \quad (3.31)$$

The expressions of $\dot{\mathbf{T}}_s(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}}$ and $\dot{\mathbf{T}}_q(\mathbf{q}) \dot{\mathbf{q}}$ are given by

$$\dot{\mathbf{T}}_s(\boldsymbol{\theta}) \dot{\boldsymbol{\theta}} = \left\{ c_1 \dot{\boldsymbol{\theta}} \dot{\boldsymbol{\theta}}^T + c_2 \tilde{\boldsymbol{\theta}} \dot{\boldsymbol{\theta}} \dot{\boldsymbol{\theta}}^T + c_3 (\boldsymbol{\theta}^T \dot{\boldsymbol{\theta}}) \boldsymbol{\theta} \dot{\boldsymbol{\theta}}^T + c_5 \left[\dot{\boldsymbol{\theta}} \dot{\boldsymbol{\theta}}^T + (\boldsymbol{\theta}^T \dot{\boldsymbol{\theta}}) \mathbf{I} \right] \right\} \dot{\boldsymbol{\theta}}, \quad (3.32)$$

where

$$\begin{aligned} c_1 &= \frac{\theta \cos \theta - \sin \theta}{\theta^3}, & c_2 &= \frac{\theta \sin \theta + 2 \cos \theta - 2}{\theta^4}, \\ c_3 &= \frac{3 \sin \theta - 2\theta - \theta \cos \theta}{\theta^5}, & c_5 &= \frac{\theta - \sin \theta}{\theta^3}, \end{aligned}$$

and

$$\dot{\mathbf{T}}_q(\mathbf{q})\dot{\mathbf{q}} = \left(\dot{\mathbf{q}}^T \dot{\mathbf{q}} + \frac{(\mathbf{q}^T \dot{\mathbf{q}})^2}{4q_o^2} \right) \frac{\mathbf{q}}{4q_o}. \quad (3.33)$$

Note that the material angular velocity and acceleration can also be obtained by similar formulations. This issue, however will not be dealt with in this thesis.

3.2 Time stepping method for finite rotations

Newmark time integration method

The standard relations of the Newmark time integration method for displacements are

$$\mathbf{u}_{n+1} = \mathbf{u}_n + h\dot{\mathbf{u}}_n + h^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\mathbf{u}}_n + \beta \ddot{\mathbf{u}}_{n+1} \right], \quad (3.34)$$

$$\dot{\mathbf{u}}_{n+1} = \dot{\mathbf{u}}_n + h \left[(1 - \gamma) \ddot{\mathbf{u}}_n + \gamma \ddot{\mathbf{u}}_{n+1} \right]. \quad (3.35)$$

Due to the non-additive property of finite rotations, the update procedure for rotational quantities needs to be carefully treated. In [55], Simo and Vu-Quoc have proposed to directly apply the classic Newmark updates to material angular velocity, material acceleration and material incremental rotational vector. The algorithm is given by

$$\Theta_{n+1} = h\dot{\Theta}_n + h^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\Theta}_n + \beta \ddot{\Theta}_{n+1} \right], \quad (3.36)$$

$$\dot{\Theta}_{n+1} = \dot{\Theta}_n + h \left[(1 - \gamma) \ddot{\Theta}_n + \gamma \ddot{\Theta}_{n+1} \right]. \quad (3.37)$$

This Newmark time integration method is adopted in [14, 32, 34, 35, 58]. Ibrahimbegović and Mikdad [30] reformulated this method using spatial forms

$$\Theta_{n+1} = h\dot{\Theta}_n + h^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\Theta}_n + \beta \mathbf{\Lambda}_{n+1}^T \ddot{\Theta}_{n+1} \right], \quad (3.38)$$

$$\dot{\Theta}_{n+1} = \mathbf{\Lambda}_{n+1} \left[\dot{\Theta}_n + h(1 - \gamma) \ddot{\Theta}_n \right] + h\gamma \ddot{\Theta}_{n+1}, \quad (3.39)$$

where $\mathbf{\Lambda}_{n+1} = \exp(\widetilde{\Theta_{n+1}})$.

As pointed out by Mäkinen [42], the Newmark update procedures, given by Eqs. (3.36) and (3.37), involve vectors that do not belong to the same tangent space. Indeed, the vectors $\dot{\mathbf{w}}_{n+1}$ and $\ddot{\mathbf{w}}_{n+1}$ lie in a tangent space different from the one containing vectors $\mathbf{\Theta}_{n+1}$, $\dot{\mathbf{w}}_n$ and $\ddot{\mathbf{w}}_n$. Hence, these update procedures are not formally correct.

Cardona and Geradin, in [11], have proposed a different way to use Newmark time integration method for finite rotations. The time derivatives of the incremental rotational vector are used instead of angular velocity and acceleration. Hence, the additive property of the spatial incremental rotational vector can be used and the classic Newmark update procedure for translations is also applied to the rotational variables. In this case, the algorithm is given by

$$\mathbf{\Theta}_{n+1} = h\dot{\mathbf{\Theta}}_n + h^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{\mathbf{\Theta}}_n + \beta \ddot{\mathbf{\Theta}}_{n+1} \right], \quad (3.40)$$

$$\dot{\mathbf{\Theta}}_{n+1} = \dot{\mathbf{\Theta}}_n + h \left[(1 - \gamma) \ddot{\mathbf{\Theta}}_n + \gamma \ddot{\mathbf{\Theta}}_{n+1} \right]. \quad (3.41)$$

Predictors for the iterative procedure

In the nonlinear dynamic analysis of structures, displacements (rotations), velocities and accelerations need to be computed at each time step. For these three unknowns, Newmark time stepping method gives only two relations. Thus one unknown must be predicted as the initial value for the solution at the time t_{n+1} . A poor predictor can increase the number of iterations and in some cases makes the procedure fail to converge.

In [38], four predictors, found in the literature, were implemented and tested:

- The first predictor (Pred. 1), called as “Unchanged displacements”, is used by Simo and Vu-Quoc [55]. The displacements and the rotations at t_n are taken as predictor for the solution at t_{n+1} .
- The second one (Pred. 2), referred to “Null accelerations”, is used by Cardona and Geradin [11], Mäkinen [42], Chung and Hulbert [12]. Zero translational and rotational accelerations are taken as predictor for the solution at t_{n+1} .
- The third one (Pred. 3), called as “Unchanged accelerations”, used by Forsell [18], proposes to take the translational and rotational accelerations at t_n as predictor for the solution at t_{n+1} .
- The last predictor (Pred. 4), proposed by Crisfield [13], uses the tangent operator at t_n to predict initial values at t_{n+1} . This predictor was first presented in [13] for the case of a linear inertia force vector and then extended in [38] for an arbitrary nonlinear inertia force vector. At t_{n+1} , the nodal displacement vector $\mathbf{d}$ is initialized

as follows

$$\mathbf{d}_{n+1}^0 = \mathbf{d}_n + \Delta \mathbf{d}. \quad (3.42)$$

For the HHT α method, $\Delta \mathbf{d}$ is calculated by

$$\begin{aligned} \mathbf{K}_{\text{Total},n} \Delta \mathbf{d} = & (1 + \alpha) \mathbf{f}_{\text{ext},n+1} - \mathbf{f}_{g,n} - \mathbf{f}_{k,n} - \alpha \mathbf{f}_{\text{ext},n} \\ & + \mathbf{C}_{k,n} \left(\frac{\gamma}{\beta} \dot{\mathbf{d}}_n - \frac{h(2\beta - \gamma)}{2\beta} \ddot{\mathbf{d}}_n \right) + \frac{\mathbf{M}_n}{\beta h^2} \left(h \dot{\mathbf{d}}_n + \frac{h^2}{2} \ddot{\mathbf{d}}_n \right), \end{aligned} \quad (3.43)$$

where

$$\mathbf{K}_{\text{Total},n} = (1 + \alpha) \mathbf{K}_{\text{Static},n} + \mathbf{K}_{\text{Dyn},n}.$$

Tangent dynamic matrix

The nonlinear equation of motion is solved using the Newton-Raphson iterative procedure. Thus, the linearization of the inertia force vector need to be derived

$$\Delta \mathbf{f}_k = \mathbf{M} \Delta \ddot{\mathbf{d}} + \mathbf{C}_k \Delta \dot{\mathbf{d}} + \mathbf{K}_k \Delta \mathbf{d}, \quad (3.44)$$

where $\mathbf{K}_k$, $\mathbf{C}_k$ and $\mathbf{M}$ respectively denote centrifugal matrix, gyroscopic matrix and mass matrix.

In several case, the full linearization of the inertia force is difficult to obtain. In order to overcome this problem and to reduce the CPU time, Geradin and Cardona [11, 21] recommended to keep only the mass matrix and to neglect the gyroscopic and centrifugal matrices, as follow

$$\Delta \mathbf{f}_k \approx \mathbf{M} \Delta \ddot{\mathbf{d}}. \quad (3.45)$$

In [38], another simplification was proposed and the following three alternatives were tested: the exact tangent inertia matrix (i.e. mass, gyroscopic and centrifugal terms), only the mass matrix (as proposed in [11, 21]) and the mass and gyroscopic terms (the new proposal). The new proposal is given by

$$\Delta \mathbf{f}_k \approx \mathbf{M} \Delta \ddot{\mathbf{d}} + \mathbf{C}_k \Delta \dot{\mathbf{d}}. \quad (3.46)$$

3.3 Comparison of dynamic formulations

Four nonlinear dynamic formulations of 3D beam elements, based on several parameterizations and Newmark time integration methods for finite rotations (see Table 3.3), were compared in [38]. For all of them, the inertia force vector and tangent inertia matrix were derived in a total Lagrangian context and computed using two Gauss points. The first formulation was proposed by Simo and Vu-Quoc [55] and used spatial spin variables. A modification of the computation of the rotational quantities at the Gauss points was introduced

in order to get a higher efficiency. The second formulation, based on the incremental rotational vector, was developed by Ibrahimbegović and Mikdad in [30]. The third one, also based on the incremental rotational vector, was proposed by Cardona and Geradin [11]. It was reformulated using the spatial form of the incremental rotational vector instead of the material one. The fourth formulation employed three of the four Euler parameters (quaternion) as rotational variables. This idea was introduced by Battini [7] for static analysis and then developed in [38] for dynamics.

Regarding the static deformational terms, i.e. the internal force vector and tangent stiffness matrix, the corotational beam elements developed by Battini and Pacoste [7, 8] were employed. For the deformational static part, the corotational formulation with the local element $t3d$ proposed in [8] was adopted. The expressions of the static deformational terms will be given in detail in Chapter 4.

Table 3.1: Four nonlinear dynamic formulations.

Formulation	Rotational variables		Newmark time integration method
Simo & Vu-Quoc [55]	Spatial spin variables		Simo & Vu-Quoc
Ibrahimbegović & Mikdad [30]	Incremental	rotational vector	
Cardona & Geradin [11]			Cardona & Geradin
New formulation [38]	Incremental parameters	Euler	

3.4 Numerical example - Free-free flexible beam with disks

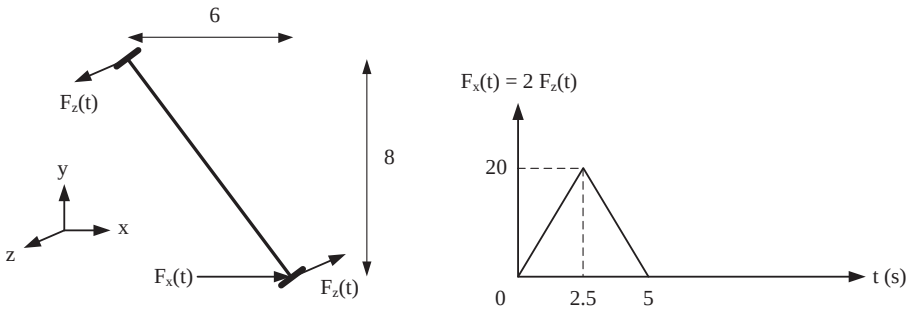


Fig. 3.2: Free-free flexible beam with disks and loading.

This example, proposed by Ibrahimbegović and Mikdad [30], analyzes the flight of a flexible beam with rigid disks attached to it. The initial configuration is given in Fig. 3.2. The mass per unit length of the beam and the inertia dyadic of the cross section in initial configuration are $A_\rho = 1$, $\mathbf{J}_\rho = \text{diag}(20, 10, 10)$, respectively. The other material properties are $EA = GA = 10^4$, $EI = GJ = 500$. The disks have a point mass $M = 10.0$ and an inertia matrix $\mathbf{J}_\rho = \text{diag}(200.0, 100.0, 100.0)$. The system is set into motion by applying a couple of out-of-plane force $F_z(t)$ and an in-plane force $F_x(t)$ with $F_x(t) = 2F_z(t)$. The beam was discretized using 10 beam elements and the disks were modeled with one-node element.

The example was analyzed using the four dynamic formulations. The nonlinear equation of motion was solved using the HHT α method with $\alpha = -0.01$. The following convergence criterion was adopted: the norm of the residual vector must be less than the prescribed tolerance $\varepsilon_f = 10^{-5}$. The time-step size $\Delta t = 0.10$ s was chosen and time histories for the displacements of the beam's lower right end were presented in Fig. 3.3. Since all formulations provided the same numerical results, the curves presented could be obtained with any of the formulations discussed.

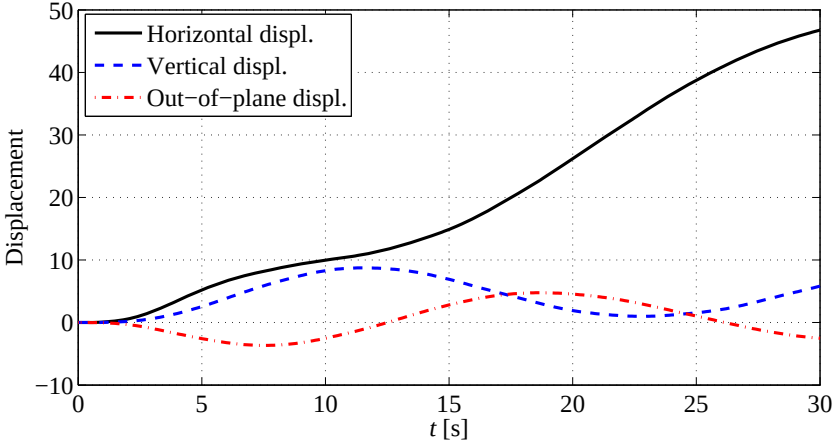


Fig. 3.3: Free-free flexible beam with disks. End responses time-history.

Choice of the predictor

The four predictors described in Section 3.2 were compared with several time-step sizes. The exact tangent inertia matrices were used. The total numbers of iterations for the whole time history and for each formulation were given in Table 3.2.

Pred. 1 converged in all the cases, but required the largest number of iterations. In fact,

Table 3.2: Number of iterations.

		Pred. 1	Pred. 2	Pred. 3	Pred. 4
Spat. Spin. Var.	$\Delta t = 0.25$ s	479	363	363	357
	$\Delta t = 0.10$ s	1181	902	901	878
Inc. Rot. V. 1	$\Delta t = 0.25$ s	479	363	363	357
	$\Delta t = 0.10$ s	1181	902	901	877
Inc. Rot. V. 2	$\Delta t = 0.25$ s	479	363	363	357
	$\Delta t = 0.10$ s	1181	902	897	877
Inc. Euler	$\Delta t = 0.25$ s	479	363	363	357
	$\Delta t = 0.10$ s	1181	902	897	877

this predictor assumes that the configuration of the structure does not change during a time step which does not happen in almost all cases.

In fact, for all the formulations, the best alternative was to use Pred. 4. The number of iterations was significantly smaller than for Pred. 1. This can be explained by the fact that Pred. 4 assumes linearity of the system during the time step. This is often a good approximation, especially with small time steps. In the sequel, only Pred. 4 was used in the numerical calculations.

Exact versus simplified dynamic tangent matrix and comparison of the four formulations

For each formulation, the exact tangent inertia matrix, the simplified matrix proposed by the author (Simpl. 1) and the one proposed by Geradin and Cardona (Simpl. 2) were tested. Table 3.3 showed the CPU time and the total number of iterations (in parentheses) for each formulation.

Table 3.3: Numerical performances.

	Spat. Spin. Var.	Inc. Rot. V. 1	Inc. Rot. V. 2	Inc. Euler
Exact	19.5 (878)	21.9 (877)	25.1 (877)	22.5 (877)
Simpl. 1	19.2 (885)	20.2 (882)	20.0 (881)	18.6 (881)
Simpl. 2	22.8 (1150)	23.8 (1140)	21.8 (1144)	21.1 (1154)

From the numerical results, it can be concluded that:

- For the first two formulations, Simpl. 2 was the slowest alternative. It increased the total number of iterations by about 25% to 38% compared with the exact matrix. In fact, for these formulations, the tangent matrices do not require a lot of CPU time, so the CPU time gained by the simplification cannot compensate for the CPU time needed for the extra iterations. From the results, it was quite difficult to make a choice between the exact dynamic tangent matrix and the first simplification. The difference in terms of CPU time cost between these two matrices were small. The author recommended to use the exact tangent matrix with these two formulations.
- For the last two formulations, the best alternative was to use Simpl. 1 which reduced the CPU time by about 25% when compared to the exact matrix and by about 15% when compared to Simpl. 2.
- The formulation using the spatial Euler parameters with Simpl. 1 was the fastest.

Influence of the time-step size

The previous results showed that the formulation using the spatial Euler parameters with Simpl. 1 was the fastest. However, the example was solved with only one time-step size. To have a more complete view, the example was repeated with different time steps. The formulations Spat. Spin. Var. and Inc. Rot. V. 1 were implemented with the exact tangent inertia matrix. The other two formulations were implemented using Simpl. 1.

The results were presented in Table 3.4. They showed that the choice of time-step size did not affect the hierarchy between the formulations.

Table 3.4: Numerical performances with various time-step size.

	Spat. Spin. Var.	Inc. Rot. V. 1	Inc. Rot. V. 2	Inc. Euler
$\Delta t = 0.1$ s	19.5 (878)	21.9 (877)	20.0 (881)	18.6 (881)
$\Delta t = 0.5$ s	5.7 (239)	5.9 (237)	5.3 (239)	4.9 (239)

Chapter 4

3D corotational beam elements with solid cross-section

The purpose of this chapter is to present a 3D corotational element for dynamic analysis of beams with solid cross-sections. The corotational framework used in this chapter is the classic one proposed by Nour-Omid and Rankin [45, 50]. The main idea of the corotational method is to decompose the motion of the element into rigid body and pure deformational parts. During the rigid body motion, a local coordinates system, attached to the element, moves and rotates with it. The deformational part is measured in this local system. The main interest of the approach is that different assumptions can be made to represent the local deformations, giving rise to different possibilities for the local element formulation.

Several local formulations were proposed by Battini and Pacoste [8], and Alsafadie et al. [1] for the geometrically and materially static nonlinear analysis of beam structures with the corotational approach. The results of a comparative study of 3D beam formulations, which can be found in [1], showed that local beam elements based on cubic interpolations were more efficient and accurate than the ones which employ linear interpolations.

For 2D dynamics, Le et al. [37] developed a consistent 2D corotational beam element for nonlinear dynamics. Cubic interpolations were used to describe the local displacements and to derive both inertia and internal terms. Numerical results demonstrated that the formulation was more efficient than the classic formulations (i.e. with the constant Timoshenko and the constant lumped mass matrices).

For 3D dynamics, the possibility to use cubic interpolations to derive inertia terms is still an issue. Crisfield et al. [14] suggested to use a constant Timoshenko mass matrix along with local cubic interpolations to derive the internal force vector and the corresponding tangent stiffness matrix. As pointed out by Crisfield et al. [14], this combination is not consistent but it provides reasonable results when the number of elements is large enough. Hsiao et al. [24] presented a corotational formulation for the nonlinear analysis of 3D

beams. However the corotational framework adopted in [24] was different from the classic one as proposed by Nour-Omid and Rankin [45] and adopted in [37]. Therefore, it is interesting to extend the consistent 2D corotational dynamic formulation presented in [37] to 3D beam structures. Such a beam element has been proposed in [39]. The novelty of this element is that the corotational framework has been used to derive not only the internal force vector and the tangent stiffness matrix but also the inertia force vector and the tangent dynamic matrix. The same cubic interpolations have been adopted to formulate both inertia and internal local terms. In doing so, the complex expressions of the inertia terms have been significantly simplified by adopting a proper approximation for the local rotations. To enhance the efficiency of the iterative procedure, the less significant term in the tangent dynamic matrix has been ignored. Four numerical examples have been investigated with the objective to compare the performances of the new formulation against two other approaches. The first approach is similar to the one presented above, but linear local interpolations, instead of cubic ones, are used to derive only the dynamic terms. The purpose is to evaluate the influence of the choice of the local interpolations on the dynamic terms. The second approach is the classic total Lagrangian formulation proposed by Simo and Vu-Quoc [53–55].

The organisation of the chapter is as follows. Section 4.1 gives a complete presentation of the corotational beam kinematics in 3D. In Section 4.2 and 4.3, the global internal force vector and the tangent stiffness matrix are derived for different parameterizations of finite rotations. The local beam formulation is then given in Section 4.4. The content of Sections 4.1-4.4 are entirely taken from the work of Battini [5, 6]. They are reproduced here in order to make this thesis self-contained. Section 4.5 briefly presents the expressions of the inertia force vector and the tangent dynamic matrix. Finally, in Section 4.6, one example, reproduced from [39], is investigated.

4.1 Beam kinematics

In this work, the corotational framework introduced by Nour-Omid and Rankin [45], and further developed by Pacoste and Eriksson [49] and Battini and Pacoste [8] is fully adopted.

The definition of the corotational two node beam element involves several coordinate systems, see Fig. 4.1. First a global reference system is defined by the triad of unit orthogonal vectors $\mathbf{e}_j$ ($j = 1, 2, 3$). Next, a local system which continuously rotates and translates with the element is selected. The orthonormal basis vectors of the local system are denoted by $\mathbf{r}_j$ ($j = 1, 2, 3$). In the initial (undeformed) configuration, the local system is defined by the orthonormal triad $\mathbf{e}_j^0$. In addition, $\mathbf{t}_j^1$ and $\mathbf{t}_j^2$ ($j = 1, 2, 3$), denote two unit triads rigidly attached to nodes 1 and 2.

According to the main idea of the corotational formulation, the motion of the element from the initial to the final deformed configuration is split into a rigid body component and a deformational part. The rigid body motion consists of a rigid translation and rotation of

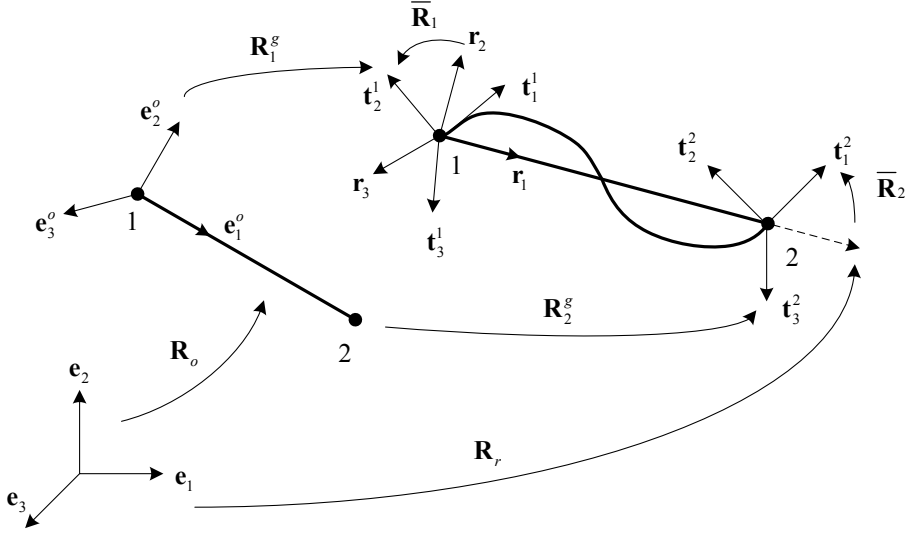


Fig. 4.1: Beam kinematics and the coordinate systems.

the local element frame. The origin of the local system is taken at node 1 and thus the rigid translation is defined by $\mathbf{u}_1^g$, the translation at node 1. Here and in the sequel, superscript g indicates quantities expressed in the global reference system. The rigid rotation is such that the new orientation of the local reference system is defined by an orthogonal matrix $\mathbf{R}_r$, given by

$$\mathbf{R}_r = [\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{r}_3]. \quad (4.1)$$

The first coordinate axis of the local system is defined by the line connecting nodes 1 and 2 of the element. Consequently, $\mathbf{r}_1$ is given by

$$\mathbf{r}_1 = \frac{\mathbf{x}_2^g + \mathbf{u}_2^g - \mathbf{x}_1^g - \mathbf{u}_1^g}{l_n}, \quad (4.2)$$

with $\mathbf{x}_i^g$ ($i = 1, 2$) denoting the nodal coordinates in the initial undeformed configuration and l_n denoting the current length of the beam, i.e.

$$l_n = \|\mathbf{x}_2^g + \mathbf{u}_2^g - \mathbf{x}_1^g - \mathbf{u}_1^g\|. \quad (4.3)$$

The remaining two axes are determined with the help of an auxiliary vector $\mathbf{p}$. In the initial configuration $\mathbf{p}$ is directed along the local $\mathbf{e}_2^o$ direction, whereas in the deformed configuration its orientation is obtained from

$$\mathbf{p} = \frac{1}{2}(\mathbf{p}_1 + \mathbf{p}_2), \quad \mathbf{p}_i = \mathbf{R}_i^g \mathbf{R}_o [0 \ 1 \ 0]^T \quad (i = 1, 2), \quad (4.4)$$

where $\mathbf{R}_i^g$ is the orthogonal matrix used to specify the orientation of the nodal triad $\mathbf{t}_j^i$, and $\mathbf{R}_o$ specifies the orientation of the local frame in the initial configuration, i.e. $\mathbf{R}_o = [\mathbf{e}_1^o \ \mathbf{e}_2^o \ \mathbf{e}_3^o]$. The unit vectors $\mathbf{r}_2$ and $\mathbf{r}_3$ are then computed by the following vector products

$$\mathbf{r}_3 = \frac{\mathbf{r}_1 \times \mathbf{p}}{\|\mathbf{r}_1 \times \mathbf{p}\|}, \quad \mathbf{r}_2 = \mathbf{r}_3 \times \mathbf{r}_1, \quad (4.5)$$

and the orthogonal matrix $\mathbf{R}_r$ in Eq. (4.1) is completely determined.

The rigid motion previously described, is accompanied by local deformational displacements and rotations with respect to the local element axes. In this context, due to the particular choice of the local system, the local translations at node 1 are zero. Moreover, at node 2, the only non zero component is the translation along $\mathbf{r}_1$. This can easily be evaluated according to

$$\bar{u} = l_n - l_o, \quad (4.6)$$

with l_o denoting the length of the beam in the original undeformed configuration. Here and in the sequel, an overbar denotes a deformational kinematic quantity.

The global rotations at node i can be expressed in terms of the rigid rotation of the local axes, defined by $\mathbf{R}_r$, followed by a local rotation relative to these axes. The latter is defined by the orthogonal matrix $\bar{\mathbf{R}}_i$. Consequently, the orientation of the nodal triad $\mathbf{t}_j^i$ can be obtained by means of the product $\mathbf{R}_r \bar{\mathbf{R}}_i$. On the other hand, (see Fig. 4.1) this orientation can also be obtained through the product $\mathbf{R}_i^g \mathbf{R}_o$, which gives

$$\bar{\mathbf{R}}_i = \mathbf{R}_r^T \mathbf{R}_i^g \mathbf{R}_o \quad (i = 1, 2). \quad (4.7)$$

The local rotations are then evaluated from

$$\bar{\boldsymbol{\theta}}_i = \log \left(\bar{\mathbf{R}}_i \right) \quad (i = 1, 2). \quad (4.8)$$

With respect to the moving frame, local (deformational) displacements $\mathbf{d}_l$ are defined by extracting the rigid body modes from the global displacements $\mathbf{d}_g^g$. Due to the choice of the local coordinate system, the local nodal displacement vector $\mathbf{d}_l$ has only seven components and is given by

$$\mathbf{d}_l = \left[\bar{u} \ \bar{\boldsymbol{\theta}}_1^T \ \bar{\boldsymbol{\theta}}_2^T \right]^T. \quad (4.9)$$

The variation of the local nodal displacement vector is

$$\delta \mathbf{d}_l = \left[\delta \bar{u} \ \delta \bar{\boldsymbol{\theta}}_1^T \ \delta \bar{\boldsymbol{\theta}}_2^T \right]^T, \quad (4.10)$$

and the global counterpart is given by

$$\delta \mathbf{d}_g = \left[\delta \mathbf{u}_1^{gT} \ \delta \mathbf{w}_1^{gT} \ \delta \mathbf{u}_2^{gT} \ \delta \mathbf{w}_2^{gT} \right]^T, \quad (4.11)$$

with $\delta \mathbf{w}_i^g$ ($i = 1, 2$) denoting spatial spin variables as defined in Eq. (3.7).

The connection between the variations of local and global displacements is defined by a transformation matrix $\mathbf{B}$

$$\delta \mathbf{d}_l = \mathbf{B} \delta \mathbf{d}_g. \quad (4.12)$$

The expression of $\mathbf{B}$ is derived using a sequence of two changes of variables, as described in the following subsections.

Change of variables $\delta \bar{\boldsymbol{\theta}} \longrightarrow \delta \bar{\mathbf{w}}$

The general procedure for evaluating the transformation matrix $\mathbf{B}$ involves the variations of Eqs. (4.6) and (4.7). Referring to Eq. (4.7), admissible variations $\delta \bar{\mathbf{R}}_i$ ($i = 1, 2$) are computed (see Eq. (3.7)) according to

$$\delta \bar{\mathbf{R}}_i = \widetilde{\delta \bar{\mathbf{w}}_i} \bar{\mathbf{R}}_i, \quad (4.13)$$

with $\delta \bar{\mathbf{w}}_i$ denoted spatial spin variables.

The local rotational vector $\delta \bar{\boldsymbol{\theta}}_i$ is defined in Eq. (4.8). Using Eqs. (3.11), (4.8) and (4.13), the change of variables from $\delta \bar{\boldsymbol{\theta}}_i$ to $\delta \bar{\mathbf{w}}_i$ is given by

$$\delta \bar{\boldsymbol{\theta}} = \mathbf{T}_s^{-1}(\bar{\boldsymbol{\theta}}) \delta \bar{\mathbf{w}}, \quad (4.14)$$

which, by introducing the notation

$$\delta \mathbf{d}_a = \begin{bmatrix} \delta \bar{u} & \delta \bar{\mathbf{w}}_1^T & \delta \bar{\mathbf{w}}_2^T \end{bmatrix}^T, \quad (4.15)$$

gives

$$\delta \mathbf{d}_l = \mathbf{B}_a \delta \mathbf{d}_a, \quad \mathbf{B}_a = \begin{bmatrix} 1 & \mathbf{0}_{[1 \times 3]} & \mathbf{0}_{[1 \times 3]} \\ \mathbf{0}_{[3 \times 1]} & \mathbf{T}_s^{-1}(\bar{\boldsymbol{\theta}}_1) & \mathbf{0} \\ \mathbf{0}_{[3 \times 1]} & \mathbf{0} & \mathbf{T}_s^{-1}(\bar{\boldsymbol{\theta}}_2) \end{bmatrix}. \quad (4.16)$$

Here and in the sequel $\mathbf{0}_{[i \times j]}$ denotes an $i \times j$ zero matrix. For a 3×3 zero matrix the notation $\mathbf{0}$ is however used.

Change of variables $\delta \mathbf{d}_a \longrightarrow \delta \mathbf{d}_g$

The second step of the variable change involves $\delta \mathbf{d}_a$ and $\delta \mathbf{d}_g$, as defined in Eqs. (4.15) and (4.11), respectively.

Referring first to the local axial translation $\bar{u}$, the variations of Eq. (4.6) give

$$\delta \bar{u} = \delta l_n = \mathbf{r} \delta \mathbf{d}_g, \quad \mathbf{r} = \begin{bmatrix} -\mathbf{r}_1^T & \mathbf{0}_{[1 \times 3]} & \mathbf{r}_1^T & \mathbf{0}_{[1 \times 3]} \end{bmatrix}. \quad (4.17)$$

For the rotational terms, the variations of Eq. (4.7) are needed

$$\delta \bar{\mathbf{R}}_i = \delta \mathbf{R}_r^T \mathbf{R}_i^g \mathbf{R}_o + \mathbf{R}_r^T \delta \mathbf{R}_i^g \mathbf{R}_o, \quad (4.18)$$

where $\delta \bar{\mathbf{R}}_i$ is defined in Eq. (4.13) whereas $\delta \mathbf{R}_i^g$ and $\delta \mathbf{R}_r$ are computed using the spatial form of Eq. (3.7), i.e.

$$\delta \mathbf{R}_i^g = \widetilde{\delta \mathbf{w}_i^g} \mathbf{R}_i^g, \quad \delta \mathbf{R}_r = \widetilde{\delta \mathbf{w}_r^g} \mathbf{R}_r. \quad (4.19)$$

$\delta \mathbf{R}_r^T$ is calculated from the orthogonality condition $\mathbf{R}_r \mathbf{R}_r^T = \mathbf{I}$ which, by differentiation and introduction of Eq. (4.19) gives

$$\delta \mathbf{R}_r \mathbf{R}_r^T + \mathbf{R}_r \delta \mathbf{R}_r^T = 0, \quad (4.20)$$

$$\widetilde{\delta \mathbf{w}_r^g} \mathbf{R}_r \mathbf{R}_r^T + \mathbf{R}_r \delta \mathbf{R}_r^T = 0, \quad (4.21)$$

and then

$$\delta \mathbf{R}_r^T = -\mathbf{R}_r^T \widetilde{\delta \mathbf{w}_r^g}. \quad (4.22)$$

Using Eqs. (4.13), (4.19) and (4.22), Eq. (4.18) can be rewritten as

$$\begin{aligned} \widetilde{\delta \bar{\mathbf{w}}_i} \bar{\mathbf{R}}_i &= -\mathbf{R}_r^T \widetilde{\delta \mathbf{w}_r^g} \mathbf{R}_i^g \mathbf{R}_o + \mathbf{R}_r^T \widetilde{\delta \mathbf{w}_i^g} \mathbf{R}_i^g \mathbf{R}_o \\ &= -\mathbf{R}_r^T \widetilde{\delta \mathbf{w}_r^g} \mathbf{R}_r \mathbf{R}_r^T \mathbf{R}_i^g \mathbf{R}_o + \mathbf{R}_r^T \widetilde{\delta \mathbf{w}_i^g} \mathbf{R}_r \mathbf{R}_r^T \mathbf{R}_i^g \mathbf{R}_o \\ &= (\widetilde{\delta \mathbf{w}_i^e} - \widetilde{\delta \mathbf{w}_r^e}) \bar{\mathbf{R}}_i, \end{aligned} \quad (4.23)$$

where use has been made of Eq. (4.7) and of the fact that $\mathbf{R}_r$ transforms a vector and a tensor from global to local coordinates according to

$$\mathbf{x}^e = \mathbf{R}_r^T \mathbf{x}^g, \quad \widetilde{\mathbf{x}}^e = \mathbf{R}_r^T \widetilde{\mathbf{x}}^g \mathbf{R}_r. \quad (4.24)$$

Thus, Eq. (4.23) gives

$$\delta \bar{\mathbf{w}}_i = \delta \mathbf{w}_i^e - \delta \mathbf{w}_r^e \quad (i = 1, 2). \quad (4.25)$$

Further, let

$$\delta \mathbf{d}_g^e = \mathbf{E}^T \delta \mathbf{d}_g, \quad \mathbf{E} = \begin{bmatrix} \mathbf{R}_r & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_r & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R}_r \end{bmatrix}. \quad (4.26)$$

Then, using the chain rule, $\delta \bar{\mathbf{w}}_i$ is evaluated as

$$\delta \bar{\mathbf{w}}_i = \frac{\partial \bar{\mathbf{w}}_i}{\partial \mathbf{d}_g^e} \frac{\partial \mathbf{d}_g^e}{\partial \mathbf{d}_g} \delta \mathbf{d}_g = \frac{\partial \bar{\mathbf{w}}_i}{\partial \mathbf{d}_g^e} \mathbf{E}^T \delta \mathbf{d}_g \quad (i = 1, 2). \quad (4.27)$$

Substituting from Eq. (4.25) gives

$$\begin{bmatrix} \delta \bar{\mathbf{w}}_1 \\ \delta \bar{\mathbf{w}}_2 \end{bmatrix} = \left(\begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix} - \begin{bmatrix} \mathbf{G}^T \\ \mathbf{G}^T \end{bmatrix} \right) \mathbf{E}^T \delta \mathbf{d}_g = \mathbf{P} \mathbf{E}^T \delta \mathbf{d}_g, \quad (4.28)$$

where the matrix $\mathbf{G}$ is defined by

$$\mathbf{G}^T = \frac{\partial \mathbf{w}_r^e}{\partial \mathbf{d}_g^e}. \quad (4.29)$$

Hence, from Eqs. (4.17) and (4.28), the connection between $\delta \mathbf{d}_a$ and $\delta \mathbf{d}_g$ is given by

$$\delta \mathbf{d}_a = \mathbf{B}_g \delta \mathbf{d}_g, \quad \mathbf{B}_g = \begin{bmatrix} \mathbf{r} \\ \mathbf{P} \mathbf{E}^T \end{bmatrix}. \quad (4.30)$$

Expression of $\mathbf{G}$

The expression of $\mathbf{G}$ is obtained from Eq. (4.19) which can be rewritten as

$$\widetilde{\delta \mathbf{w}_r^g} = \delta \mathbf{R}_r \mathbf{R}_r^T. \quad (4.31)$$

and after the transformation (4.24) as

$$\widetilde{\delta \mathbf{w}_r^e} = \mathbf{R}_r^T \delta \mathbf{R}_r. \quad (4.32)$$

From Eq. (4.1) and the above equation, it can easily be found that

$$\delta \mathbf{w}_r^e = \begin{bmatrix} \delta w_{r_1}^e \\ \delta w_{r_2}^e \\ \delta w_{r_3}^e \end{bmatrix} = \begin{bmatrix} -\mathbf{r}_2^T \delta \mathbf{r}_3 \\ -\mathbf{r}_3^T \delta \mathbf{r}_1 \\ \mathbf{r}_2^T \delta \mathbf{r}_1 \end{bmatrix}. \quad (4.33)$$

Introducing the notation $\mathbf{u}_i^g = [u_{i1}^g \ u_{i2}^g \ u_{i3}^g]^T$ ($i = 1, 2$), differentiation of Eq. (4.2) gives

$$\delta \mathbf{r}_1^g = \frac{1}{l_n} [\mathbf{I} - \mathbf{r}_1 \mathbf{r}_1^T] \begin{bmatrix} \delta u_{21}^g - \delta u_{11}^g \\ \delta u_{22}^g - \delta u_{12}^g \\ \delta u_{23}^g - \delta u_{13}^g \end{bmatrix}, \quad (4.34)$$

and after transformation (4.24) in the local coordinate system, it is obtained

$$\delta \mathbf{r}_1^e = \frac{1}{l_n} \begin{bmatrix} \delta u_{21}^e - \delta u_{11}^e \\ \delta u_{22}^e - \delta u_{12}^e \\ \delta u_{23}^e - \delta u_{13}^e \end{bmatrix}. \quad (4.35)$$

Hence, since the local expressions of $\mathbf{r}_2$ and $\mathbf{r}_3$ are $[0 \ 1 \ 0]^T$ and $[0 \ 0 \ 1]^T$, Eq. (4.33) gives

$$\delta w_{r_2}^e = \frac{1}{l_n} (\delta u_{13}^e - \delta u_{23}^e), \quad (4.36)$$

$$\delta w_{r_3}^e = \frac{1}{l_n} (\delta u_{22}^e - \delta u_{12}^e). \quad (4.37)$$

The evaluation of $\delta w_{r_1}^e$ is more complicated and can be performed as follows. Differentiation of Eq. (4.4) gives

$$\begin{aligned}\delta \mathbf{p} &= \frac{1}{2}(\delta \mathbf{R}_1^g + \delta \mathbf{R}_2^g) \mathbf{R}_o [0 \ 1 \ 0]^T \\ &= \frac{1}{2}(\widetilde{\delta \mathbf{w}_1^g} \mathbf{R}_1^g + \widetilde{\delta \mathbf{w}_2^g} \mathbf{R}_2^g) \mathbf{R}_o [0 \ 1 \ 0]^T \\ &= \frac{1}{2}(\widetilde{\delta \mathbf{w}_1^g} \mathbf{p}_1 + \widetilde{\delta \mathbf{w}_2^g} \mathbf{p}_2).\end{aligned}\quad (4.38)$$

The local expressions of the vectors $\mathbf{p}$, $\mathbf{p}_1$ and $\mathbf{p}_2$ are denoted by

$$\mathbf{R}_r^T \mathbf{p} = \begin{bmatrix} p_1 \\ p_2 \\ 0 \end{bmatrix}, \quad \mathbf{R}_r^T \mathbf{p}_1 = \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \end{bmatrix}, \quad \mathbf{R}_r^T \mathbf{p}_2 = \begin{bmatrix} p_{21} \\ p_{22} \\ p_{23} \end{bmatrix}. \quad (4.39)$$

The last coordinate of $\mathbf{R}_r^T \mathbf{p}$ is zero since $\mathbf{p}$ is perpendicular to $\mathbf{r}_3$.

The local expression of $\delta \mathbf{p}$ can be deduced from Eq. (4.38) as

$$\delta \mathbf{p}^e = \frac{1}{2} \widetilde{\delta \mathbf{w}_1^e} \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \end{bmatrix} + \frac{1}{2} \widetilde{\delta \mathbf{w}_2^e} \begin{bmatrix} p_{21} \\ p_{22} \\ p_{23} \end{bmatrix}, \quad (4.40)$$

which after calculation gives

$$\delta \mathbf{p}^e = \frac{1}{2} \begin{bmatrix} -p_{12} \delta w_{13}^e + p_{13} \delta w_{12}^e - p_{22} \delta w_{23}^e + p_{23} \delta w_{22}^e \\ + p_{11} \delta w_{13}^e - p_{13} \delta w_{11}^e + p_{21} \delta w_{23}^e - p_{23} \delta w_{21}^e \\ - p_{11} \delta w_{12}^e + p_{12} \delta w_{11}^e - p_{21} \delta w_{22}^e + p_{22} \delta w_{21}^e \end{bmatrix}. \quad (4.41)$$

The following notations are introduced

$$\eta = \frac{p_1}{p_2}, \quad \eta_{11} = \frac{p_{11}}{p_2}, \quad \eta_{12} = \frac{p_{12}}{p_2}, \quad \eta_{21} = \frac{p_{21}}{p_2}, \quad \eta_{22} = \frac{p_{22}}{p_2}. \quad (4.42)$$

The differentiation of $\mathbf{r}_3$ is calculated from its definition (4.5). By noting that $\|\mathbf{r}_1 \times \mathbf{p}\| = p_2$, the first line of Eq. (4.33) can be rewritten as

$$\delta w_{r_1}^e = -\frac{\mathbf{r}_2^T}{p_2} \delta \mathbf{r}_1^e \times \mathbf{p} - \frac{\mathbf{r}_2^T}{p_2} \mathbf{r}_1 \times \delta \mathbf{p}^e - \delta \left(\frac{1}{p_2} \right) \mathbf{r}_2^T (\mathbf{r}_1 \times \mathbf{p}). \quad (4.43)$$

The last term in the above equation is zero. The two other terms can be evaluated from Eqs. (4.35), (4.39) and (4.41). The result, after some work, is

$$\delta w_{r_1}^e = \frac{\eta}{l_n} (\delta w_1^e - \delta w_2^e) - \frac{\eta_{11}}{2} \delta w_{12}^e + \frac{\eta_{12}}{2} \delta w_{11}^e - \frac{\eta_{21}}{2} \delta w_{22}^e + \frac{\eta_{22}}{2} \delta w_{21}^e. \quad (4.44)$$

Finally, the expression for the matrix $\mathbf{G}$ is

$$\mathbf{G}^T = \begin{bmatrix} 0 & 0 & \frac{\eta}{l_n} & \frac{\eta_{12}}{2} & -\frac{\eta_{11}}{2} & 0 & 0 & 0 & -\frac{\eta}{l_n} & \frac{\eta_{22}}{2} & -\frac{\eta_{21}}{2} & 0 \\ 0 & 0 & \frac{1}{l_n} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{l_n} & 0 & 0 & 0 \\ 0 & -\frac{1}{l_n} & 0 & 0 & 0 & 0 & 0 & \frac{1}{l_n} & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (4.45)$$

4.2 Global internal force vector and tangent stiffness matrix

The expression of the internal force vector in global coordinates $\mathbf{f}_g$, can be obtained by equating the internal virtual work in both the global and local systems

$$V = \delta \mathbf{d}_l^T \mathbf{f}_l = \delta \mathbf{d}_g^T \mathbf{f}_g. \quad (4.46)$$

Substituting Eq. (4.12) into the previous, provides the result of interest

$$\mathbf{f}_g = \mathbf{B}^T \mathbf{f}_l. \quad (4.47)$$

The local internal force vector is defined as

$$\mathbf{f}_l = \begin{bmatrix} n & \mathbf{m}_1^T & \mathbf{m}_2^T \end{bmatrix}^T, \quad (4.48)$$

where n denotes the axial force whereas $\mathbf{m}_1$ and $\mathbf{m}_2$ denote the moments at nodes 1 and 2, respectively.

The expression of the tangent stiffness matrix in global coordinates $\mathbf{K}_g$, is obtained by taking the variations of Eq. (4.47), which gives

$$\mathbf{K}_g = \mathbf{B}^T \mathbf{K}_l \mathbf{B} + \left. \frac{\partial(\mathbf{B}^T \mathbf{f}_l)}{\partial \mathbf{d}_g} \right|_{\mathbf{f}_l}. \quad (4.49)$$

In Eqs. (4.47) and (4.49), $\mathbf{B}$ and $\partial(\mathbf{B}^T \mathbf{f}_l)/\partial \mathbf{d}_g$ play the role of transformation matrices required in order to re-express $\mathbf{f}_l$ and $\mathbf{K}_l$ in global coordinates. These matrices, which actually define the corotational framework, depend on the nonlinear functions in Eq. (4.46) and thus on the choice of the local coordinate system. However, they are independent of the particular strain definition used in order to derive $\mathbf{f}_l$ and $\mathbf{K}_l$. Consequently, various corotational elements defined using different local strain assumptions but the same type of local coordinate system will share the same transformation matrices, i.e. the corotational formulation is “element independent” [45]. Using this property, various local assumptions can be placed at the core of the corotational formulation and tested for efficiency and accuracy.

As shown in Section 4.2, the change of variables from $\mathbf{d}_l$ to $\mathbf{d}_g$ must be performed in two steps, therefore the transformations in Eqs. (4.47) and (4.49) are also derived using these two steps, as described in the following.

A virtual work equation gives

$$\mathbf{f}_a = \mathbf{B}_a^T \mathbf{f}_l, \quad (4.50)$$

with $\mathbf{f}_a$ denoting the internal force vector consistent with $\delta \mathbf{d}_a$. The corresponding transformation for the local tangent stiffness matrices, i.e. $\mathbf{K}_l$ and $\mathbf{K}_a$, is obtained by taking the variation of Eq. (4.50)

$$\delta \mathbf{f}_a = \mathbf{B}_a^T \delta \mathbf{f}_l + \delta \mathbf{B}_a^T \mathbf{f}_l, \quad (4.51)$$

where, by definition

$$\delta \mathbf{f}_l = \mathbf{K}_l \delta \mathbf{d}_l, \quad \delta \mathbf{f}_a = \mathbf{K}_a \delta \mathbf{d}_a. \quad (4.52)$$

Using Eqs. (4.16), (4.51) and (4.52) gives the required transformation

$$\mathbf{K}_a = \mathbf{B}_a^T \mathbf{K}_l \mathbf{B}_a + \mathbf{K}_h, \quad \mathbf{K}_h = \begin{bmatrix} 0 & \mathbf{0}_{[1 \times 3]} & \mathbf{0}_{[1 \times 3]} \\ \mathbf{0}_{[3 \times 1]} & \mathbf{K}_{h1} & \mathbf{0} \\ \mathbf{0}_{[3 \times 1]} & \mathbf{0} & \mathbf{K}_{h2} \end{bmatrix}. \quad (4.53)$$

The expressions of $\mathbf{K}_{h1}$ and $\mathbf{K}_{h2}$ are computed from

$$\frac{\partial}{\partial \bar{\mathbf{w}}} [\mathbf{T}_s^{-T} \mathbf{x}] = \frac{\partial}{\partial \bar{\boldsymbol{\theta}}} [\mathbf{T}_s^{-T} \mathbf{x}] \frac{\partial \bar{\boldsymbol{\theta}}}{\partial \bar{\mathbf{w}}} = \frac{\partial}{\partial \bar{\boldsymbol{\theta}}} [\mathbf{T}_s^{-T} \mathbf{x}] \mathbf{T}_s^{-1}, \quad (4.54)$$

with the vector $\mathbf{x}$ maintained constant during differentiation. Using Eq. (3.7) gives after some algebra

$$\frac{\partial}{\partial \bar{\mathbf{w}}} [\mathbf{T}_s^{-T} \mathbf{x}] = \left[\eta [\bar{\boldsymbol{\theta}} \mathbf{x}^T - 2 \mathbf{x} \bar{\boldsymbol{\theta}}^T + (\bar{\boldsymbol{\theta}}^T \cdot \mathbf{x}) \mathbf{I}] + \mu \bar{\boldsymbol{\theta}}^2 [\mathbf{x} \bar{\boldsymbol{\theta}}^T] - \frac{1}{2} \tilde{\mathbf{x}} \right] \mathbf{T}_s^{-1}(\bar{\boldsymbol{\theta}}), \quad (4.55)$$

with the coefficients η and μ given by

$$\eta = \frac{2 \sin \bar{\theta} - \bar{\theta} (1 + \cos \bar{\theta})}{2 \bar{\theta}^2 \sin \bar{\theta}}, \quad \mu = \frac{\bar{\theta} (\bar{\theta} + \sin \bar{\theta}) - 8 \sin^2(\bar{\theta}/2)}{4 \bar{\theta}^4 \sin^2(\bar{\theta}/2)}. \quad (4.56)$$

Thus, $\mathbf{K}_{hi}$ ($i = 1, 2$) are evaluated from Eq. (4.55) with $\bar{\boldsymbol{\theta}} = \bar{\boldsymbol{\theta}}_i$ and $\mathbf{x} = \mathbf{m}_i$, with $\mathbf{m}_i$ as defined in Eq. (4.48).

Using Eq. (4.30), the internal force vector in global coordinates is computed as

$$\mathbf{f}_g = \mathbf{B}_g^T \mathbf{f}_a. \quad (4.57)$$

Note that, according to the sequence of variable changes previously defined, the matrix $\mathbf{B}$ in Eq. (4.46) is explicitly given by the product $\mathbf{B}_a \mathbf{B}_g$.

Differentiation of Eq. (4.57) gives

$$\delta \mathbf{f}_g = \mathbf{B}_g^T \mathbf{K}_a \mathbf{B}_g \delta \mathbf{d}_g + \delta \mathbf{r}^T f_{a1} + \delta(\mathbf{E} \mathbf{P}^T) \mathbf{m}, \quad (4.58)$$

with

$$\mathbf{m} = \begin{bmatrix} f_{a2} & f_{a3} & f_{a4} & f_{a5} & f_{a6} & f_{a7} \end{bmatrix}^T, \quad (4.59)$$

where f_{a_i} ($i = 1..7$) denotes the i^{th} component of the vector $\mathbf{f}_a$.

From Eqs. (4.17) and (4.34), it can easily be derived that

$$\delta \mathbf{r}^T = \mathbf{D} \delta \mathbf{d}_g, \quad \mathbf{D} = \begin{bmatrix} \mathbf{D}_3 & \mathbf{0} & -\mathbf{D}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -\mathbf{D}_3 & \mathbf{0} & \mathbf{D}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \mathbf{D}_3 = \frac{1}{l_n} (\mathbf{I} - \mathbf{r}_1 \mathbf{r}_1^T). \quad (4.60)$$

The last term in expression (4.58) is evaluated from

$$\delta(\mathbf{E} \mathbf{P}^T) \mathbf{m} = \delta \mathbf{E} \mathbf{P}^T \mathbf{m} + \mathbf{E} \delta \mathbf{P}^T \mathbf{m}. \quad (4.61)$$

By introducing

$$\mathbf{P}^T \mathbf{m} = \begin{bmatrix} \mathbf{n}_1 \\ \mathbf{m}_1 \\ \mathbf{n}_2 \\ \mathbf{m}_2 \end{bmatrix}, \quad (4.62)$$

and using Eqs. (4.26) and (4.32), the first term in Eq. (4.61) can be expressed as

$$\delta \mathbf{E} \mathbf{P}^T \mathbf{m} = \begin{bmatrix} \mathbf{R}_r \widetilde{\delta \mathbf{w}_r^e} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_r \widetilde{\delta \mathbf{w}_r^e} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R}_r \widetilde{\delta \mathbf{w}_r^e} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R}_r \widetilde{\delta \mathbf{w}_r^e} \end{bmatrix} \begin{bmatrix} \mathbf{n}_1 \\ \mathbf{m}_1 \\ \mathbf{n}_2 \\ \mathbf{m}_2 \end{bmatrix} = \mathbf{E} \begin{bmatrix} \widetilde{\delta \mathbf{w}_r^e} \mathbf{n}_1 \\ \widetilde{\delta \mathbf{w}_r^e} \mathbf{m}_1 \\ \widetilde{\delta \mathbf{w}_r^e} \mathbf{n}_2 \\ \widetilde{\delta \mathbf{w}_r^e} \mathbf{m}_2 \end{bmatrix}, \quad (4.63)$$

which, using the relation

$$\widetilde{\mathbf{a}} \mathbf{b} = -\widetilde{\mathbf{b}} \mathbf{a}, \quad (4.64)$$

gives

$$\delta \mathbf{E} \mathbf{P}^T \mathbf{m} = -\mathbf{E} \mathbf{Q} \delta \mathbf{w}_r^e, \quad \mathbf{Q} = \begin{bmatrix} \widetilde{\mathbf{n}}_1 \\ \widetilde{\mathbf{m}}_1 \\ \widetilde{\mathbf{n}}_2 \\ \widetilde{\mathbf{m}}_2 \end{bmatrix}. \quad (4.65)$$

Then, by using Eq. (4.45), it is obtained

$$\delta \mathbf{E} \mathbf{P}^T \mathbf{m} = -\mathbf{E} \mathbf{Q} \mathbf{G}^T \mathbf{E}^T \delta \mathbf{d}_g. \quad (4.66)$$

The calculation of the second term of Eq. (4.61) requires the value of $\delta \mathbf{P}^T$ which can be obtained by introducing the matrix $\mathbf{A}$ such as

$$\mathbf{A}^T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -l_n & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & l_n & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (4.67)$$

and by noting that

$$\mathbf{A}^T \mathbf{G} = \mathbf{I}. \quad (4.68)$$

Differentiation of the above equation gives

$$\delta \mathbf{A}^T \mathbf{G} + \mathbf{A}^T \delta \mathbf{G} = \mathbf{0}, \quad (4.69)$$

and hence

$$\delta \mathbf{G} = -\mathbf{A}^{-T} \delta \mathbf{A}^T \mathbf{G} = -\mathbf{G} \delta \mathbf{A}^T \mathbf{G}. \quad (4.70)$$

Further, using the definition of $\mathbf{P}$ in Eq. (4.28) gives

$$\delta \mathbf{P} = -\mathbf{C} \delta \mathbf{G}^T, \quad \mathbf{C} = \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \end{bmatrix}, \quad (4.71)$$

which can be rewritten as

$$\delta \mathbf{P}^T = -\delta \mathbf{G} \mathbf{C}^T = \mathbf{G} \delta \mathbf{A}^T \mathbf{G} \mathbf{C}^T. \quad (4.72)$$

Then, the second term of Eq. (4.61) becomes

$$\mathbf{E} \delta \mathbf{P}^T \mathbf{m} = \mathbf{E} \mathbf{G} \delta \mathbf{A}^T \mathbf{G} \mathbf{C}^T \mathbf{m}, \quad (4.73)$$

which can be simplified after symbolic matrix multiplications as

$$\mathbf{E} \delta \mathbf{P}^T \mathbf{m} = \mathbf{E} \mathbf{G} \mathbf{a} \delta l_n, \quad (4.74)$$

with

$$\mathbf{a} = \begin{bmatrix} 0 \\ \frac{\eta}{l_n} (f_{a_2} + f_{a_5}) - \frac{1}{l_n} (f_{a_3} + f_{a_6}) \\ \frac{1}{l_n} (f_{a_4} + f_{a_7}) \end{bmatrix}. \quad (4.75)$$

Introducing Eq. (4.17) in Eq. (4.74) gives

$$\mathbf{E} \delta \mathbf{P}^T \mathbf{m} = \mathbf{E} \mathbf{G} \mathbf{a} \mathbf{r} \delta \mathbf{d}_g. \quad (4.76)$$

Finally, from Eqs. (4.58), (4.61), (4.66) and (4.76) the expression of the global tangent stiffness matrix is

$$\mathbf{K}_g = \mathbf{B}_g^T \mathbf{K}_a \mathbf{B}_g + \mathbf{K}_m, \quad \mathbf{K}_m = \mathbf{D} f_{a_1} - \mathbf{E} \mathbf{Q} \mathbf{G}^T \mathbf{E}^T + \mathbf{E} \mathbf{G} \mathbf{a} \mathbf{r}. \quad (4.77)$$

4.3 Finite rotation parameters

When the spin variables are chosen to represent the finite rotations, the global internal force vector and the tangent stiffness matrix are calculated as given in previous section. When another parametrization of the finite rotations is used, a change of variables is required. In this section, such changes of variables for spatial incremental rotational vector and spatial incremental Euler parameters are presented. These two parameterizations were presented in Section 3.1 and used for the second and the third formulations in [38].

Spatial incremental rotational vector

Let $\mathbf{d}_r^g$ denote the following vector of global nodal displacements

$$\mathbf{d}_r^g = \begin{bmatrix} \mathbf{u}_1^{gT} & \boldsymbol{\theta}_1^{gT} & \mathbf{u}_2^{gT} & \boldsymbol{\theta}_2^{gT} \end{bmatrix}^T, \quad (4.78)$$

where $\boldsymbol{\theta}_i^g$ denotes the spatial incremental rotational vector at node i . The change to the new kinematic variables in $\mathbf{d}_r^g$ requires the connection between $\delta \mathbf{d}_g$ as defined in Eq. (4.11) and $\delta \mathbf{d}_r^g$. This connection can easily be constructed using Eq. (3.8)

$$\delta \mathbf{d}_g = \mathbf{B}_r \delta \mathbf{d}_r^g, \quad \mathbf{B}_r = \begin{bmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_s(\boldsymbol{\theta}_1^g) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{T}_s(\boldsymbol{\theta}_2^g) \end{bmatrix}. \quad (4.79)$$

The global internal force vector $\mathbf{f}_r$ and tangent stiffness matrix $\mathbf{K}_r$, consistent with $\mathbf{d}_r^g$, are then given by

$$\mathbf{f}_r = \mathbf{B}_r^T \mathbf{f}_g, \quad \mathbf{K}_r = \mathbf{B}_r^T \mathbf{K}_g \mathbf{B}_r + \mathbf{K}_v, \quad (4.80)$$

where

$$\mathbf{K}_v = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{v1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{K}_{v2} \end{bmatrix}. \quad (4.81)$$

The expressions of $\mathbf{K}_{v1}$ and $\mathbf{K}_{v2}$ are obtained from

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\theta}} [\mathbf{T}_s^T \mathbf{x}] &= - \left(\frac{\sin \theta}{\theta} - \left(\frac{\sin(\theta/2)}{(\theta/2)} \right)^2 \right) (\mathbf{n} \times \mathbf{x}) \mathbf{n}^T + \frac{1}{2} \left(\frac{\sin(\theta/2)}{(\theta/2)} \right)^2 \tilde{\mathbf{x}} \\ &+ \left(\cos \theta - \frac{\sin \theta}{\theta} \right) \frac{1}{\theta} [\mathbf{x} \mathbf{n}^T - (\mathbf{n}^T \mathbf{x}) \mathbf{n} \mathbf{n}^T] \\ &+ \left(1 - \frac{\sin \theta}{\theta} \right) \frac{1}{\theta} [\mathbf{n} \mathbf{x}^T - 2(\mathbf{n}^T \mathbf{x}) \mathbf{n} \mathbf{n}^T + (\mathbf{n}^T \mathbf{x}) \mathbf{I}], \end{aligned} \quad (4.82)$$

with the vector $\mathbf{x}$ maintained constant during the differentiation.

Thus, $\mathbf{K}_{v1}$ and $\mathbf{K}_{v2}$ are evaluated from Eq. (4.82) with $\boldsymbol{\theta} = \boldsymbol{\theta}_1^g$ and $\boldsymbol{\theta} = \boldsymbol{\theta}_2^g$, respectively and $\mathbf{x} = [f_{g4} \ f_{g5} \ f_{g6}]^T$, $\mathbf{x} = [f_{g10} \ f_{g11} \ f_{g12}]^T$, respectively.

Spatial incremental Euler parameters

The transformation matrices required in this second case are constructed on the basis of Eq. (3.13) which provides the connection between $\delta \mathbf{w}^g$ and $\delta \mathbf{q}^g$. Following a procedure entirely similar to that used in the preceding subsection, the vector $\mathbf{d}_r^g$ is now defined as

$$\mathbf{d}_r^g = \left[\mathbf{u}_1^{gT} \ \mathbf{q}_1^{gT} \ \mathbf{u}_2^{gT} \ \mathbf{q}_2^{gT} \right]^T, \quad (4.83)$$

where $\mathbf{q}_i^g$ denotes the incremental Euler parameters at node i . The transformation equations for the internal force vector and the tangent stiffness matrix are formally identical to Eqs. (4.79)-(4.81), by replacing $\mathbf{T}_s(\boldsymbol{\theta}_i^g)$ with $\mathbf{T}_q(\mathbf{q}_i^g)$.

In this case, the matrices $\mathbf{K}_{vi}$ are evaluated with $\mathbf{q} = \mathbf{q}_i^g$ ($i = 1, 2$) and $\mathbf{x} = [f_{g4} \ f_{g5} \ f_{g6}]^T$, $\mathbf{x} = [f_{g10} \ f_{g11} \ f_{g12}]^T$, respectively, from

$$\mathbf{K}_{vi} = \frac{1}{16 q_o^3} \mathbf{H}. \quad (4.84)$$

The components of $\mathbf{H}$ are

$$\begin{aligned} A &= (q_1 x_1 + q_2 x_2 + q_3 x_3), \quad H_{11} = (4 q_o^2 + q_1^2) A, \\ H_{22} &= (4 q_o^2 + q_2^2) A, \quad H_{33} = (4 q_o^2 + q_3^2) A, \\ H_{12} &= 4 q_o^2 (q_1 x_2 - q_2 x_1 - q_o x_3) + q_1 q_2 A, \\ H_{13} &= 4 q_o^2 (q_1 x_3 - q_3 x_1 + q_o x_2) + q_1 q_3 A, \\ H_{21} &= 4 q_o^2 (q_2 x_1 - q_1 x_2 + q_o x_3) + q_2 q_1 A, \\ H_{23} &= 4 q_o^2 (q_2 x_3 - q_3 x_2 - q_o x_1) + q_2 q_3 A, \\ H_{31} &= 4 q_o^2 (q_3 x_1 - q_1 x_3 - q_o x_2) + q_3 q_1 A, \\ H_{32} &= 4 q_o^2 (q_3 x_2 - q_2 x_3 + q_o x_1) + q_3 q_2 A, \end{aligned} \quad (4.85)$$

where x_j ($j = 1, 2, 3$) are components of $\mathbf{x}$.

4.4 Local internal force vector and tangent stiffness matrix

The purpose of this section is to define the internal force vector $\mathbf{f}_l$ and tangent stiffness matrix $\mathbf{K}_l$ in local coordinates.

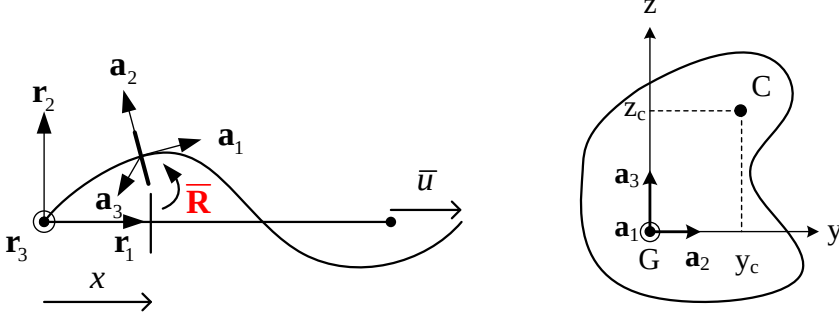


Fig. 4.2: Local beam configuration.

Assuming elastic material behavior, $\mathbf{f}_l$ and $\mathbf{K}_l$ can be derived from the strain energy Φ expressed as a function of the local displacements $\mathbf{d}_l$, through successive differentiations

$$\mathbf{f}_l = \frac{\partial \Phi}{\partial \mathbf{d}_l}, \quad \mathbf{K}_l = \frac{\partial^2 \Phi}{\partial \mathbf{d}_l^2}. \quad (4.86)$$

As a general rule, all kinematic quantities are referred to the local element frame $\mathbf{r}_i$ ($i = 1, 2, 3$), as defined in Eqs. (4.2), (4.4) and (4.5). The origin of the local system is taken at node 1 with $\mathbf{r}_1$ directed along the line of centroids. However $\mathbf{r}_2$ and $\mathbf{r}_3$ are not necessarily directed along the principal axes of the cross-section.

With respect to the local system, the Green-Lagrange strain components which contribute to the strain energy of the beam are given by

$$\begin{aligned} \varepsilon_{11} &= u_{1,x} + \frac{1}{2} u_{1,x}^2 + \frac{1}{2} u_{2,x}^2 + \frac{1}{2} u_{3,x}^2, \\ 2\varepsilon_{12} &= u_{1,y} + u_{2,x} + u_{1,x} u_{1,y} + u_{2,x} u_{2,y} + u_{3,x} u_{3,y}, \\ 2\varepsilon_{13} &= u_{1,z} + u_{3,x} + u_{1,x} u_{1,z} + u_{2,x} u_{2,z} + u_{3,x} u_{3,z}, \end{aligned} \quad (4.87)$$

where u_1, u_2, u_3 are the local (deformational) displacements of the current point P. In the above equation, a comma followed by an index denotes differentiation with respect to the corresponding variable.

The kinematic model proposed by Gruttmann et al. [22] is adopted, see Fig. 4.2. This kinematic description can be used for an arbitrary cross-section, therefore the local beam kinematics given here stands also for beam with arbitrary thin-walled cross-section presented in Chapter 5.

Let $\mathbf{x}_p^o(x, y, z)$ denote the position vector of point P in the initial (i.e. rotated but still undeformed) configuration and let $\mathbf{x}_p(x, y, z)$ denote the position vector of P in the current

configuration (see Fig. 4.2). These two vectors are given by

$$\begin{aligned}\mathbf{x}_P^o(x, y, z) &= \mathbf{x}_G^o(x) + y\mathbf{r}_2 + z\mathbf{r}_3, \\ \mathbf{x}_P(x, y, z) &= \mathbf{x}_G(x) + y\mathbf{a}_2(x) + z\mathbf{a}_3(x) + \alpha(x)\bar{\omega}(y, z)\mathbf{a}_1(x),\end{aligned}\quad (4.88)$$

with $\mathbf{x}_G^o$ and $\mathbf{x}_G$ denoting the position vectors of G in the initial and current configurations, respectively.

The warping function $\bar{\omega}(y, z)$ is defined within the Saint-Venant torsion theory and refers to the centroid G, i.e.

$$\bar{\omega} = \omega - y_c z + z_c y, \quad (4.89)$$

where ω refers to the shear center C with coordinates y_c, z_c . Note that in connection with ω , the following normality conditions hold

$$\int_A \omega dA = 0, \quad \int_A \omega y dA = 0, \quad \int_A \omega z dA = 0. \quad (4.90)$$

The orthonormal triad $\mathbf{a}_i$ ($i = 1, 2, 3$) which specifies the orientation of the current cross-section, is given by

$$\mathbf{a}_i = \bar{\mathbf{R}}\mathbf{r}_i, \quad (4.91)$$

where the orthogonal matrix $\bar{\mathbf{R}}$ defines a rotation relative to the local element axes, see Eq. (4.7). Using Eq. (3.4), a second order approximation of $\bar{\mathbf{R}}$ can be constructed as

$$\bar{\mathbf{R}} = \mathbf{I} + \tilde{\bar{\boldsymbol{\theta}}} + \frac{1}{2}\tilde{\bar{\boldsymbol{\theta}}}^2, \quad \tilde{\bar{\boldsymbol{\theta}}} = \begin{bmatrix} 0 & -\bar{\theta}_3 & \bar{\theta}_2 \\ \bar{\theta}_3 & 0 & -\bar{\theta}_1 \\ -\bar{\theta}_2 & \bar{\theta}_1 & 0 \end{bmatrix}. \quad (4.92)$$

Introducing Eq. (4.90) into Eq. (4.88), a second order approximation of the displacement vector $\bar{\mathbf{u}}_P = [u_1 \ u_2 \ u_3]^T = \mathbf{x}_P - \mathbf{x}_P^o$ can be evaluated as

$$\begin{aligned}u_1 &= \bar{u}_1 - y\bar{\theta}_3 + z\bar{\theta}_2 + \frac{1}{2}y\bar{\theta}_1\bar{\theta}_2 + \frac{1}{2}z\bar{\theta}_1\bar{\theta}_3 + \bar{\omega}\alpha, \\ u_2 &= \bar{u}_2 - z\bar{\theta}_1 - \frac{1}{2}y(\bar{\theta}_1^2 + \bar{\theta}_3^2) + \frac{1}{2}z\bar{\theta}_2\bar{\theta}_3 + \bar{\omega}\alpha\bar{\theta}_3, \\ u_3 &= \bar{u}_3 + y\bar{\theta}_1 - \frac{1}{2}z(\bar{\theta}_1^2 + \bar{\theta}_2^2) + \frac{1}{2}y\bar{\theta}_2\bar{\theta}_3 - \bar{\omega}\alpha\bar{\theta}_2,\end{aligned}\quad (4.93)$$

where $\bar{u}_1, \bar{u}_2, \bar{u}_3$ are the displacements of the centroid G, i.e. the components of the vector $\mathbf{x}_G - \mathbf{x}_G^o$.

Using Eqs. (4.89), (4.90) and (4.93), a second order approximation of the Green-Lagrange strains defined in Eq. (4.86) can be evaluated. At this level, two additional simplifications are introduced. First, the term $1/2\bar{u}_{1,x}^2$ in the expression of ε_{11} is neglected. Second, the

nonlinear strain components generated by warping are omitted since warping effects are rationally taken into account in a linearised way only [23]. Using these simplifications gives

$$\begin{aligned}\varepsilon_{11} &= \varepsilon_c + yk_2 + zk_3 + \frac{1}{2}r^2\bar{\theta}_{1,x}^2 + \omega\alpha_{,x}, \\ 2\varepsilon_{12} &= \gamma_{12} + \bar{\omega}_{,y}\alpha - zk_1, \\ 2\varepsilon_{13} &= \gamma_{13} + \bar{\omega}_{,z}\alpha + yk_1,\end{aligned}\tag{4.94}$$

with

$$\begin{aligned}r^2 &= y^2 + z^2, \\ \varepsilon_c &= u_{1,x} + \frac{1}{2}(u_{2,x}^2 + u_{3,x}^2), \\ k_2 &= -\bar{\theta}_{3,x} + \frac{1}{2}(\bar{\theta}_{1,x}\bar{\theta}_2 + \bar{\theta}_1\bar{\theta}_{2,x}) + u_{3,x}\bar{\theta}_{1,x} + z_c\alpha_{,x}, \\ k_3 &= \bar{\theta}_{2,x} + \frac{1}{2}(\bar{\theta}_{1,x}\bar{\theta}_3 + \bar{\theta}_1\bar{\theta}_{3,x}) - u_{2,x}\bar{\theta}_{1,x} - y_c\alpha_{,x}, \\ k_1 &= \bar{\theta}_{1,x} + \frac{1}{2}(\bar{\theta}_{2,x}\bar{\theta}_3 - \bar{\theta}_2\bar{\theta}_{3,x}), \\ \gamma_{12} &= u_{2,x} - \bar{\theta}_3 + \frac{1}{2}\bar{\theta}_1\bar{\theta}_2 + u_{3,x}\bar{\theta}_1 - u_{1,x}\bar{\theta}_3, \\ \gamma_{13} &= u_{3,x} + \bar{\theta}_2 + \frac{1}{2}\bar{\theta}_1\bar{\theta}_3 - u_{2,x}\bar{\theta}_1 + u_{1,x}\bar{\theta}_2.\end{aligned}\tag{4.95}$$

The strain energy can then be evaluated from

$$\Phi = \int_{l_o} \Phi_A dx = \int_{l_o} \left(\frac{1}{2}E \int_A \varepsilon_{11}^2 dA + \frac{1}{2}G \int_A [(2\varepsilon_{12})^2 + (2\varepsilon_{13})^2] dA \right) dx.\tag{4.96}$$

Integration through the cross-section gives

$$\begin{aligned}\int_A \varepsilon_{11}^2 dA &= A\varepsilon_c^2 + I_{yy}k_2^2 + I_{zz}k_3^2 + \frac{1}{4}I_{rr}\bar{\theta}_{1,x}^4 + I_\omega\alpha_{,x}^2 + I_o\varepsilon_c\bar{\theta}_{1,x}^2 \\ &\quad + 2I_{yz}k_2k_3 + I_{yr}k_2\bar{\theta}_{1,x}^2 + I_{zr}k_3\bar{\theta}_{1,x}^2 + I_{\omega r}\alpha_{,x}\bar{\theta}_{1,x},\end{aligned}\tag{4.97}$$

$$\begin{aligned}\int_A [(2\varepsilon_{12})^2 + (2\varepsilon_{13})^2] dA &= A(\gamma_{12}^2 + \gamma_{13}^2) + I_o k_1^2 + (I_o - J)\alpha^2 \\ &\quad - 2(I_o - J)\alpha k_1,\end{aligned}\tag{4.98}$$

where J denotes the Saint-Venant torsion modulus

$$J = \int_A [y(\bar{\omega}_{,z} + y) - z(\bar{\omega}_{,y} - z)] dA,\tag{4.99}$$

and the other section quantities are defined by

$$\begin{aligned} I_{yy} &= \int_A y^2 dA, & I_{zz} &= \int_A z^2 dA, & I_{yz} &= \int_A yz dA, \\ I_{yr} &= \int_A y r^2 dA, & I_{zr} &= \int_A z r^2 dA, & I_{rr} &= \int_A (y^2 + z^2)^2 dA, \\ I_{\omega} &= \int_A \omega^2 dA, & I_{\omega r} &= \int_A \omega r^2 dA, & I_o &= \int_A (y^2 + z^2) dA. \end{aligned} \quad (4.100)$$

In deriving Eqs. (4.98) and (4.99), the following relations have been used (cf. the proof in [22])

$$\int_A \bar{\omega}_{,y} dA = \int_A \bar{\omega}_{,z} dA = 0, \quad (4.101)$$

$$\int_A (\bar{\omega}_{,y}^2 + \bar{\omega}_{,z}^2) dA = - \int_A (\bar{\omega}_{,z} y - \bar{\omega}_{,y} z) dA = I_o - J. \quad (4.102)$$

If the Bernoulli assumption is adopted, the bending shear strains γ_{12} and γ_{13} in Eq. (4.95) are neglected, which gives

$$\bar{\theta}_2 = -u_{3,x} + \frac{1}{2} u_{2,x} \bar{\theta}_1 + u_{1,x} u_{3,x}, \quad \bar{\theta}_3 = u_{2,x} + \frac{1}{2} u_{3,x} \bar{\theta}_1 - u_{1,x} u_{2,x}. \quad (4.103)$$

By introducing the above equations into Eq. (4.95), the bending curvatures and torsional twist can be rewritten as

$$\begin{aligned} k_1 &= \bar{\theta}_{1,x} + \frac{1}{2} (u_{3,x} u_{2,xx} - u_{2,x} u_{3,xx}), \\ k_2 &= -u_{2,xx} + u_{1,xx} u_{2,x} + u_{1,x} u_{2,xx} - u_{3,xx} \bar{\theta}_1 + z_c \alpha_{,x}, \\ k_3 &= -u_{3,xx} + u_{1,xx} u_{3,x} + u_{1,x} u_{3,xx} + u_{2,xx} \bar{\theta}_1 - y_c \alpha_{,x}. \end{aligned} \quad (4.104)$$

Since the bending shear stresses are neglected, it appears logical to neglect the bi-shear F_{ω} also. Thus

$$F_{\omega} = \frac{\partial \Phi_A}{\partial \alpha} = 0, \quad (4.105)$$

where Φ_A is defined in Eq. (4.96). Introducing (4.97) and (4.98) into the above expression, gives

$$\alpha = k_1. \quad (4.106)$$

The integral (4.98) is then simplified under the form

$$\int_A [(2\varepsilon_{12})^2 + (2\varepsilon_{13})^2] dA = J k_1^2. \quad (4.107)$$

In order to avoid membrane locking, ε_{11} in Eq.(4.94) is rewritten as

$$\varepsilon_{11} = \varepsilon_{av} + y k_2 + z k_3 + \frac{1}{2} \left(r^2 - \frac{I_o}{A} \right) \bar{\theta}_{1,x} + \omega \alpha_{,x}, \quad (4.108)$$

where the “average” value ε_{av} is given by

$$\varepsilon_{av} = \frac{1}{l_o} \int_{l_o} \left[u_{1,x} + \frac{1}{2} \left(u_{2,x}^2 + u_{3,x}^2 + \frac{I_o}{A} \bar{\theta}_{1,x}^2 \right) \right] dx. \quad (4.109)$$

This gives

$$\begin{aligned} \int_A \varepsilon_{11}^2 dA = & A \varepsilon_{av}^2 + I_{yy} k_2^2 + I_{zz} k_3^2 + \frac{1}{4} \left(I_{rr} - \frac{I_o^2}{A} \right) \bar{\theta}_{1,x}^4 + I_{\omega} \alpha_{,x}^2 \\ & + 2 I_{yz} k_2 k_3 + I_{yr} k_2 \bar{\theta}_{1,x}^2 + I_{zr} k_3 \bar{\theta}_{1,x}^2 + I_{\omega r} \alpha_{,x} \bar{\theta}_{1,x}^2. \end{aligned} \quad (4.110)$$

Beams with solid cross-sections

For such beams, the cross-section is assumed to be bi-symmetric, with $\mathbf{r}_2$ and $\mathbf{r}_3$ directed along the principal axes of the cross-section. These assumptions give

$$I_{yz} = I_{yr} = I_{zr} = I_{\omega r} = y_c = z_c = 0. \quad (4.111)$$

The warping stresses are neglected and consequently the bi-shear F_{ω} and the bi-moment M_{ω} are set to zero

$$F_{\omega} = \frac{\partial \Phi_A}{\partial \alpha} = 0, \quad M_{\omega} = \frac{\partial \Phi_A}{\partial \alpha_{,x}} = 0. \quad (4.112)$$

Introducing Eqs. (4.97) and (4.98) in the above expressions, gives

$$\alpha = k_1, \quad I_{\omega} \alpha_{,x} = 0, \quad (4.113)$$

which are further substituted in the corresponding expressions of the strain energy.

In the second paper [38], the Timoshenko element *t3d*, based on Eqs. (4.95), (4.97) and (4.98), was used. The element adopts linear interpolations for all variables and a one point Gauss rule to avoid the shear locking. In numerical examples in the paper, the section quantity I_{rr} was neglected. Extensive numerical testing showed that this simplification has no effect on the accuracy of the results.

In the third paper [39], the Bernoulli element *b3d*, based on Eqs. (4.104), (4.107), (4.109) and (4.110), was used. This element adopts linear interpolations for the axial displacement and axial rotation, cubic interpolations for the transverse displacements. Then, due to the particular choice of the local degrees of freedom, one has

$$\begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \\ \bar{\theta}_1 \\ \bar{\theta}_2 \\ \bar{\theta}_3 \end{bmatrix} = \begin{bmatrix} N_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & N_3 & 0 & 0 & N_4 \\ 0 & 0 & -N_3 & 0 & 0 & -N_4 & 0 \\ 0 & N_1 & 0 & 0 & N_2 & 0 & 0 \\ 0 & 0 & N_5 & 0 & 0 & N_6 & 0 \\ 0 & 0 & 0 & N_5 & 0 & 0 & N_6 \end{bmatrix} \mathbf{d}_l, \quad (4.114)$$

with

$$\begin{aligned} N_1 &= 1 - \frac{x}{l_o}, \quad N_2 = 1 - N_1, \quad N_3 = x \left(1 - \frac{x}{l_o}\right)^2, \\ N_4 &= -\left(1 - \frac{x}{l_o}\right) \frac{x^2}{l_o}, \quad N_5 = \left(1 - \frac{3x}{l_o}\right) \left(1 - \frac{x}{l_o}\right), \quad N_6 = \left(\frac{3x}{l_o} - 2\right) \frac{x}{l_o}. \end{aligned}$$

Let $\mathbf{u}_l = [0 \ \bar{u}_2 \ \bar{u}_3]^T$ denotes the local transverse displacement vector. This vector is calculated by

$$\mathbf{u}_l = \mathbf{P}_1 \begin{bmatrix} \bar{\theta}_1 \\ \bar{\theta}_2 \end{bmatrix}, \quad \mathbf{P}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & N_3 & 0 & 0 & N_4 \\ 0 & -N_3 & 0 & 0 & -N_4 & 0 \end{bmatrix}. \quad (4.115)$$

The local rotation of the cross-section is evaluated as follows

$$\bar{\boldsymbol{\theta}} = \mathbf{P}_2 \begin{bmatrix} \bar{\theta}_1 \\ \bar{\theta}_2 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} N_1 & 0 & 0 & N_2 & 0 & 0 \\ 0 & N_5 & 0 & 0 & N_6 & 0 \\ 0 & 0 & N_5 & 0 & 0 & N_6 \end{bmatrix}. \quad (4.116)$$

Both element types were implemented by neglecting the nonlinear terms in the expressions of the curvatures k_1, k_2, k_3 and bending shear strains γ_{12}, γ_{13} . Extensive numerical testing showed that this simplification had no effect whatsoever on the accuracy of the results; i.e. including these terms generates differences of less than 0.01% in the final results. On the other hand, certain tests showed that the Wagner term $\frac{1}{2} r^2 \bar{\theta}_{1,x}^2$ in the expression of ϵ_{11} can not be neglected or even be replaced by its average value $(\frac{1}{2} \frac{l_o}{A} \bar{\theta}_{1,x}^2)$ over the cross-section.

The Maple codes of these local beam elements, taken from the work of Battini [5], are given as follows

Maple code of *t3d* element

```
>with(linalg):
>du:=u12/L:
>t1:=(t11+t12)/2:
```

```

>t2:=(t21+t22)/2:
>t3:=(t31+t32)/2:
>dt1:=(t12-t11)/L:
>dt2:=(t22-t21)/L:
>dt3:=(t32-t31)/L:
>g12:=-t3:
>g13:=t2:
>k1:=dt1:
>k2=-dt3:
>k3:=dt2:
>Phi1:=A*du^2+Iyy*k2^2+Izz*k3^2+Irr/4*dt1^4+Io*du*dt1^2:
>Phi2:=A*(g12^2+g13^2)+J*k1^2:
>Phi:=1/2*L*(E*Phi1+G*Phi2):
>f1:=grad(Phi,[u,t11,t21,t31,t12,t22,t32]):
>k1:=hessian(Phi,[u,t11,t21,t31,t12,t22,t32]):

```

Maple code of *b3d* element

```

>with(linalg):
>u21:=0:
>u31:=0:
>u22:=0:
>u32:=0:
>f1:=1-3*(x/L)^2+2*(x/L)^3:
>f2:=x*(1-x/L)^2:
>f3:=1-f1:
>f4:=x^2*(x/L-1)/L:
>u2:=f1*u21+f2*t31+f3*u22+f4*t32:
>du2:=diff(u2,x):
>ddu2:=diff(du2,x):
>u3:=f1*u31-f2*t21+f3*u32-f4*t22:
>du3:=diff(u3,x):
>ddu3:=diff(du3,x):
>t1:=(1-x/L)*t11+x/L*t12:
>dt1:=diff(t1,x):
>k1:=dt1:
>k2=-ddu2:
>k3=-ddu3:
>eav:=u12/L+1/2/L*int(du2^2+du3^2+Io/A*dt1^2,x=0..L):
>Phi1:=A*eav^2+Iyy*k2^2+Izz*k3^2+1/4*(Irr-Io^2/A)*dt1^4:
>Phi:=1/2*int(E*Phi1+G*J*k1^2,x=0..L):
>f1:=grad(Phi,[u,t11,t21,t31,t12,t22,t32]):
>k1:=hessian(Phi,[u,t11,t21,t31,t12,t22,t32]):

```

4.5 Inertia force vector and tangent dynamic matrix

In this section, the derivation of the inertia force vector and the tangent dynamic matrix presented in [39] is briefly summarized. The novelty of the approach is that the same corotational kinematics description is adopted to formulate both inertia and internal terms, which ensures the consistency of the formulation. As already mentioned, this 3D corotational dynamic formulation is the extension of the 2D one proposed in [37].

Using the cross-section centroid as the reference point, the kinetic energy of a spatial beam element is given by

$$K = \frac{1}{2} \int_{l_0} \{ \dot{\mathbf{u}}^T A_\rho \dot{\mathbf{u}} + \dot{\mathbf{w}}^T \mathbf{I}_\rho \dot{\mathbf{w}} \} dl, \quad (4.117)$$

where $\dot{\mathbf{u}}$, $\dot{\mathbf{w}}$ are the translational and angular velocity of the cross-section and $\mathbf{I}_\rho$ is the spatial inertia dyadic. Please note that in order to simplify the notations, in this section, the superscripts “g” which refer to global quantities are dropped for $\mathbf{u}$, $\mathbf{w}$ and their derivatives.

The variation of the kinetic energy can be expressed as (see [21])

$$\delta K = - \int_{l_0} \left\{ \delta \mathbf{u}^T A_\rho \ddot{\mathbf{u}} + \delta \mathbf{w}^T \left[\mathbf{I}_\rho \ddot{\mathbf{w}} + \tilde{\mathbf{w}} \mathbf{I}_\rho \ddot{\mathbf{w}} \right] \right\} dl. \quad (4.118)$$

The inertia force vector is then derived from the following relation

$$\delta K = -\mathbf{f}_k^T \delta \mathbf{d}_g. \quad (4.119)$$

Translational displacement variables

The position of the cross-section centroid in the global coordinate system is given by

$$\mathbf{OG} = \mathbf{x}_1^g + \mathbf{u}_1^g + (x + \bar{u}_1) \mathbf{r}_1 + \bar{u}_2 \mathbf{r}_2 + \bar{u}_3 \mathbf{r}_3. \quad (4.120)$$

Using Eqs. (4.2), (4.6) and (4.114), the variation of the above equation is obtained

$$\delta \mathbf{OG} = \delta \mathbf{u} = \mathbf{N} \delta \mathbf{d}_g + \mathbf{R}_r \delta \mathbf{u}_l + \delta \mathbf{R}_r \mathbf{u}_l, \quad (4.121)$$

with $\mathbf{N} = \begin{bmatrix} N_1 \mathbf{I} & \mathbf{0} & N_2 \mathbf{I} & \mathbf{0} \end{bmatrix}$.

In the corotational approach, the local rotations at the nodes $\bar{\boldsymbol{\theta}}_i$ ($i = 1, 2$), defined in Eq. (4.8), are small and the operator $\mathbf{T}_s(\bar{\boldsymbol{\theta}}_i)$ is close to the identity matrix. Consequently, see Eq. (3.8), the following approximation is adopted

$$\delta \bar{\mathbf{w}}_i \approx \delta \bar{\boldsymbol{\theta}}_i. \quad (4.122)$$

It can be observed that for the internal force vector, this simplification, which implies to neglect the first change of variables in Section 4.2, cannot be performed. Using Eqs. (4.28) and (4.122), the differentiation of Eq. (4.115) is obtained as

$$\delta \mathbf{u}_l \approx \mathbf{P}_1 \begin{bmatrix} \delta \bar{\mathbf{w}}_1 \\ \delta \bar{\mathbf{w}}_2 \end{bmatrix} = \mathbf{P}_1 \mathbf{P} \mathbf{E}^T \delta \mathbf{d}_g. \quad (4.123)$$

Finally, the expression (4.121) can be rewritten in a more compact form as

$$\delta \mathbf{u} = \mathbf{R}_r \mathbf{H}_1 \mathbf{E}^T \delta \mathbf{d}_g, \quad (4.124)$$

with

$$\mathbf{H}_1 = \mathbf{N} + \mathbf{P}_1 \mathbf{P} - \tilde{\mathbf{u}}_r \mathbf{G}^T. \quad (4.125)$$

Finite rotation variables

Using Eqs. (4.24) and (4.25), the spatial spin variables, associated to the global rotation of a cross-section, are evaluated using

$$\delta \mathbf{w} = \mathbf{R}_r \delta \mathbf{w}^e = \mathbf{R}_r (\delta \mathbf{w}_r^e + \delta \bar{\mathbf{w}}). \quad (4.126)$$

Combining the approximation defined in Eq. (4.122) with Eqs. (4.116), (4.28) and (4.29), the spatial spin variables are computed from the global nodal displacement vector by

$$\delta \mathbf{w} = \mathbf{R}_r \mathbf{H}_2 \mathbf{E}^T \delta \mathbf{d}_g, \quad (4.127)$$

where

$$\mathbf{H}_2 = \mathbf{P}_2 \mathbf{P} + \mathbf{G}^T. \quad (4.128)$$

Inertia force vector and tangent dynamic matrix

By inserting the expressions of $\delta \mathbf{u}$ and $\delta \mathbf{w}$ given in Eqs. (4.124) and (4.127) into Eq. (4.118), the inertia force vector is obtained as

$$\mathbf{f}_k = \mathbf{E} \left[\int_{l_o} \left\{ \mathbf{H}_1^T \mathbf{R}_r^T A_\rho \ddot{\mathbf{u}} + \mathbf{H}_2^T \mathbf{R}_r^T \left[\mathbf{I}_\rho \ddot{\mathbf{w}} + \tilde{\mathbf{w}} \mathbf{I}_\rho \dot{\mathbf{w}} \right] \right\} dl \right]. \quad (4.129)$$

Linearization of this force vector is evaluated as follows

$$\Delta \mathbf{f}_k = \mathbf{M} \Delta \ddot{\mathbf{d}}_g + \mathbf{C}_k \Delta \dot{\mathbf{d}}_g + \mathbf{K}_k \Delta \mathbf{d}_g. \quad (4.130)$$

Some authors proposed to keep only the mass matrix $\mathbf{M}$, and to eliminate the gyroscopic $\mathbf{C}_k$ and centrifugal $\mathbf{K}_k$ dynamic matrices [11, 24]. However, in [38], extensive numerical studies showed that it is better to keep also the gyroscopic matrix in order to enhance the computational efficiency. The same approach was used in [39]. The centrifugal matrix, whose derivation is complicated and would give a lengthy mathematical expression, was neglected. Therefore, the iterative scheme of the present formulation was implemented with the following approximative linearization

$$\Delta \mathbf{f}_k \approx \mathbf{M} \Delta \ddot{\mathbf{d}}_g + \mathbf{C}_k \Delta \dot{\mathbf{d}}_g. \quad (4.131)$$

The expressions of the mass matrix $\mathbf{M}$ and of the gyroscopic matrix $\mathbf{C}_k$ are given by

$$\mathbf{M} = \mathbf{E} \left\{ \int_{l_0} \left(\mathbf{H}_1^T A_\rho \mathbf{H}_1 + \mathbf{H}_2^T \mathbf{I}_\rho^e \mathbf{H}_2 \right) dl \right\} \mathbf{E}^T = \mathbf{E} \mathbf{M}^e \mathbf{E}^T, \quad (4.132)$$

$$\begin{aligned} \mathbf{C}_k = \mathbf{E} \left\{ \int_{l_0} \mathbf{H}_1^T A_\rho (\mathbf{C}_1 + \mathbf{C}_3) dl + \int_{l_0} \mathbf{H}_2^T \mathbf{I}_\rho^e (\mathbf{C}_2 + \mathbf{C}_4) dl \right. \\ \left. + \int_{l_0} \mathbf{H}_2^T \left(\widetilde{\mathbf{w}}^e \mathbf{I}_\rho^e - \mathbf{I}_\rho^e \widetilde{\mathbf{w}}^e \right) \mathbf{H}_2 dl \right\} \mathbf{E}^T = \mathbf{E} \mathbf{C}_k^e \mathbf{E}^T. \end{aligned} \quad (4.133)$$

4.6 Numerical example - Right-angle cantilever beam

This classic example, introduced by Simo and Vu-Quoc [55], has often been used in order to validate various nonlinear dynamic formulations [24, 30, 35, 43]. A right-angle cantilever, depicted in Fig. 4.3, is subjected to an out-of-plane concentrated load applied at the elbow. Each member of the cantilever has a length $L = 10$. The material properties are $GA = EA = 10^6$, $GJ = EI = 10^3$. The mass per unit length of the beam and the inertia dyadic of the cross section in the initial configuration are $A_\rho = 1$ and $\mathbf{J}_\rho = \text{diag}(20, 10, 10)$, respectively.

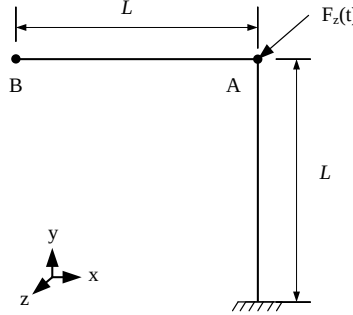


Fig. 4.3: Right-angle cantilever beam : geometrical data.

The load (see Fig. 4.4) acts only during $0 \leq t(s) \leq 2$ and thereafter the cantilever undergoes large-scale free vibrations. The resulting amplitude of vibration is of the same order of magnitude as the structure's dimensions. The purpose of this dynamic example is to assess both the cubic and the linear corotational formulations. The first one employs cubic interpolation for all terms whereas the second uses cubic interpolation for the internal force vector and the corresponding stiffness matrix, the inertia terms being evaluated considering linear interpolation. It can be observed that linear local interpolations imply $\mathbf{u}_l = \mathbf{0}_{3 \times 1}$, which simplifies significantly the computations.

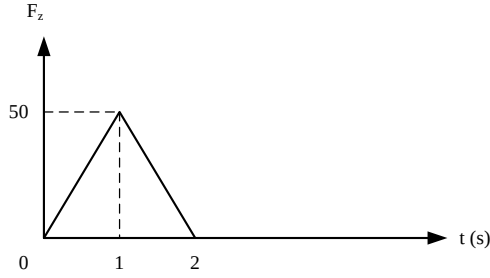
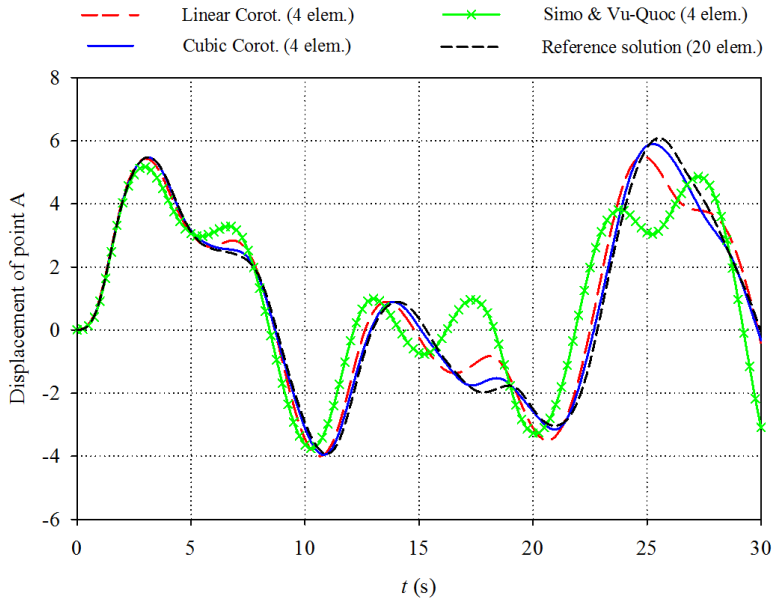


Fig. 4.4: Right-angle cantilever beam - Loading history.

The predictions of these formulations are compared against the total Lagrangian formulation proposed by Simo and Vu-Quoc [53–55]. In this formulation which employs spatial spin variables, the internal force vector and the tangent stiffness matrix are numerically integrated using one Gauss point whereas the inertia force vector and the dynamic tangent matrix are numerically integrated using two Gauss points. All the three formulations are implemented using Matlab. The same levels of optimisation in forming the element matrices and force vectors are carefully checked.

The solutions given by the three dynamic formulations are compared against a reference

Fig. 4.5: Right-angle cantilever beam - Time evolution of the displacement u_z of point A.

solution. The latter is obtained by increasing the number of elements until the three dynamic formulations converge towards the same results.

The HHT α method [13] with $\alpha = -0.05$ was employed. Regarding the predictor to initiate the iterative procedure, Pred. 4 (see Section 3.2) was adopted. This predictor was the most effective among the four ones compared in [38].

Damping was not considered and the following convergence criterion was used: the norm of the residual vector must be less than the prescribed tolerance $\varepsilon_f = 10^{-5}$.

The reference solution was obtained with a total of 20 beam elements (10 elements per member) whereas the predictions of the three aforementioned formulations were calculated using only 4 elements (2 elements per member). The time evolution of the displacement u_z of node A was depicted in Fig. 4.5. The computations were performed with $\Delta t = 0.25$ s. It can be seen that with only 4 beam elements, the cubic corotational approach gave almost the same results as the reference solution over the whole time domain. The results obtained with the linear corotational approach were very close to the reference solution with minor differences. However, the results obtained with the formulation of Simo and Vu-Quoc deviated significantly from the reference solution which is obviously due to the

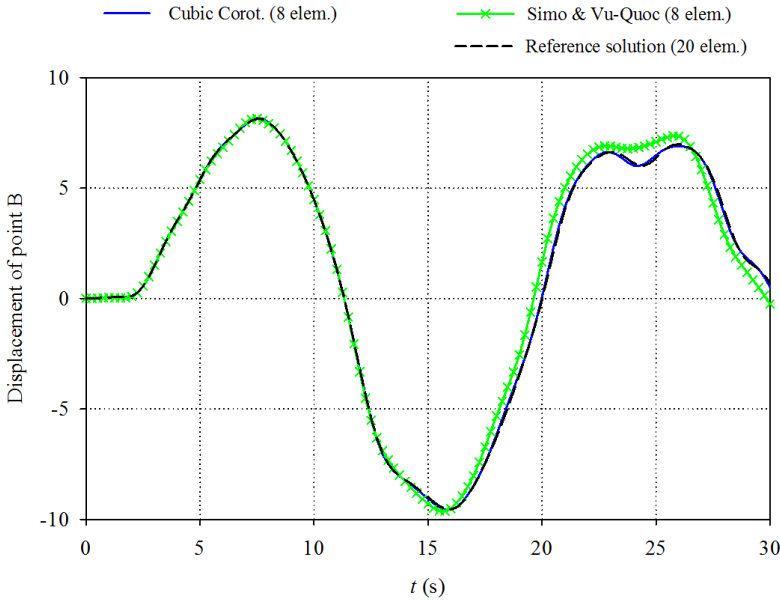


Fig. 4.6: Right-angle cantilever beam - Time evolution of the displacement u_z of point B.

linear nature of the interpolation.

The time evolution of the displacement u_z of node B, calculated with a total of 8 beam elements (4 elements per member), was depicted in Fig. 4.6. It can be noted that the differences between the formulation of Simo and Vu-Quoc and the reference solution became smaller, but also that the cubic corotational formulation gave very accurate results.

In the current example, the energy blow-up could happen at approximately $t = 45$ s. This numerical instability was reported by Jelenić and Crisfield [34, 35] and by Zupan et al. [58]. Therefore, to assess the stability of the HHT α method when combined with the new cubic formulation, this example was run for a time period of 150 s using a total of 20 beam elements and a time step $\Delta t = 0.25$ s. The time evolution of the total energy was depicted in Fig. 4.7. The result indicated that the new formulation was stable during the time period under consideration. The dissipation of energy between $t = 2$ s and $t = 30$ s was about 3%.

The CPU time and the total number of iterations required for each formulation were presented in Table 4.1. With the same number of elements, the linear corotational approach was the fastest approach and the cubic corotational approach was the slowest one. This was expected since several inertia terms disappear when local linear interpolations are adopted. Moreover, in the linear corotational approach, the inertia force vector and the tangent dynamic matrix were integrated using two Gauss points, whereas the dynamic terms of the cubic corotational one were integrated using three Gauss points.

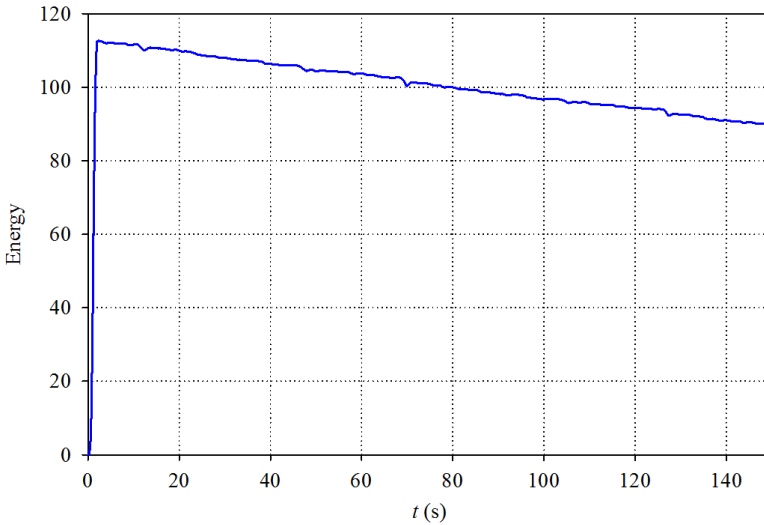


Fig. 4.7: Right-angle cantilever beam - Time evolution of the total energy.

Table 4.1: Right-angle cantilever beam - CPU time (Total number of iterations).

Number of beam elements	Linear corot.	Cubic corot.	Simo & Vu-Quoc
4	3.66 (633)	5.31 (637)	4.59 (521)
20	19.84 (636)	25.61 (612)	22.90 (517)

This example shows that with coarse meshes, Simo and Vu-Quoc's two-noded formulation and the linear corotational one give less accurate results compared to the cubic corotational approach which provides very accurate results. This demonstrates the importance of an accurate description of the deformations in the dynamic terms, which is not possible with linear interpolations. With fine meshes, all formulations give the same results.

Chapter 5

3D corotational beam elements with thin-walled open cross-section

Dynamic analysis of flexible structures undergoing large displacements and finite rotations is an attractive research topic and a large number of beam finite formulations have been proposed in the literature. Many of them have been formulated in the total Lagrangian context [4, 10, 11, 20, 30, 32, 34, 35, 41, 43, 55, 57, 58], and in the corotational context [9, 14–17, 19, 24–26, 33, 37, 39, 44, 46, 56]. However, the number of formulations which deal with nonlinear dynamics of beams with arbitrary thin-walled cross-sections is very limited [25].

In [40], a new dynamic formulation for nonlinear analysis of beams with arbitrary thin-walled cross-sections was proposed. In fact, this new formulation is the extension of the one developed in [39] by taking into account the warping deformations and the eccentricity of the shear center. Hence, the corotational method was used to derive both internal and inertia terms. For local beam description, linear interpolation was adopted for the axial displacement, whereas cubic interpolations were used for the transverse displacements and for the axial rotation.

In order to introduce the warping deformations, the kinematic description proposed by Gruttmann et al. [22] was adopted to derive both internal and inertia force vectors. Consequently, the beam element had seven degrees of freedom at each node. Moreover, several additional terms were introduced in the expressions of the inertia force vectors and the tangent dynamic matrix. The contributions of these terms in the performance of the formulation were then investigated in the numerical examples. Regarding the static deformational terms, i.e. the internal force vector and tangent stiffness matrix, the corotational beam element developed by Battini and Pacoste [8], and presented in Chapter 4, was adopted. However, in order to introduce the bending shear deformations, the cubic Hermitian functions were modified as suggested in the Interdependent Interpolation Element (IIE) formulation [51]. Five numerical examples were analysed to evaluate the accuracy of the formulation

against Abaqus 3D solid elements.

The relevant points of this new formulation is summarized in this chapter. In Section 5.1, the expression of the kinetic energy, incorporating the warping deformations and the eccentricity of the shear center is presented. Section 5.2 describes the corotational beam kinetics for thin-walled beam, while the local beam description is given in Section 5.3. The internal force vector and the tangent stiffness matrix, taken from [5], is introduced in Section 5.4. The inertia force vector and the tangent dynamic matrix is briefly presented in Section 5.5. Finally, one numerical example, taken from [40], is investigated. The purpose of this example is to assess the performance of the present formulation, and also to highlight the influence of the eccentricity of the shear center on the inertia terms.

5.1 Kinetic energy

The kinematic description proposed by Gruttmann et al. [22] is adopted (see Fig. 5.1). A beam with an arbitrary cross-section is considered. G and C are the centroid and the

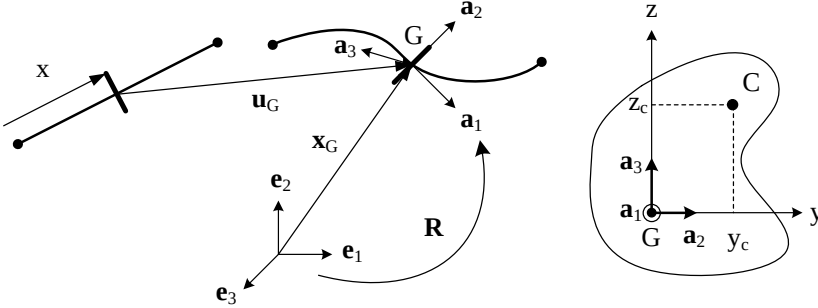


Fig. 5.1: Initial and current configuration of the beam.

shear center of the cross-section. $\mathbf{e}_i$ and $\mathbf{a}_i$ ($i = 1, 2, 3$) denote the global and cross-section-attached orthogonal coordinates systems, respectively. The transformation from the global coordinates system to the cross-section-attached one is defined by

$$\mathbf{a}_i = \mathbf{R} \mathbf{e}_i. \quad (5.1)$$

Let $\mathbf{x}_P(x, y, z)$ denote the position vector of an arbitrary point P in the current configuration

$$\mathbf{x}_P(x, y, z) = \mathbf{x}_G(x) + y\mathbf{a}_2(x) + z\mathbf{a}_3(x) + \alpha(x)\bar{\omega}(y, z)\mathbf{a}_1(x), \quad (5.2)$$

with $\mathbf{x}_G$ denoting the position vector of G in the current configuration.

The warping function $\bar{\omega}(y, z)$ is defined within the Saint-Venant torsion theory and refers to the centroid G (see Eqs. (4.89) and (4.90)).

Using Eq. (5.1), the expression (5.2) can be put in the form

$$\mathbf{x}_p = \mathbf{x}_G + \mathbf{R} (\mathbf{X} + \alpha \mathbf{n}_{\bar{\omega}}), \quad (5.3)$$

with $\mathbf{X} = [0 \ y \ z]^T$ and $\mathbf{n}_{\bar{\omega}} = [\bar{\omega} \ 0 \ 0]^T$.

By taking the time derivative of the above expression and using Eq. (3.23), the velocity is obtained as

$$\dot{\mathbf{u}}_p = \dot{\mathbf{u}}_G + \mathbf{R} (-\tilde{\mathbf{X}} \dot{\boldsymbol{\omega}} - \alpha \tilde{\mathbf{n}}_{\bar{\omega}} \dot{\boldsymbol{\omega}} + \dot{\alpha} \mathbf{n}_{\bar{\omega}}). \quad (5.4)$$

The kinetic energy of the beam is then calculated by

$$K = \frac{1}{2} \int_V \rho \dot{\mathbf{u}}_p^T \dot{\mathbf{u}}_p \, dV. \quad (5.5)$$

After some calculations, the spatial form is obtained as

$$K = \frac{1}{2} \int_{l_0} [\dot{\mathbf{u}}_G^T A \rho \dot{\mathbf{u}}_G + \dot{\mathbf{w}}^T \mathbf{I}_\rho \dot{\mathbf{w}}] \, dl + \frac{1}{2} \int_{l_0} [\dot{\mathbf{w}}^T \mathbf{I}_\alpha \dot{\mathbf{w}} - 2 \dot{\mathbf{w}}^T \mathbf{I}_b \dot{\alpha} + J_{\bar{\omega}} \dot{\alpha}^2] \, dl, \quad (5.6)$$

where

$$\mathbf{I}_\rho = \mathbf{R} \mathbf{J}_\rho \mathbf{R}^T, \quad \mathbf{I}_{\bar{\omega}} = \mathbf{R} \mathbf{J}_{\bar{\omega}} \mathbf{R}^T, \quad \mathbf{I}_\alpha = \mathbf{R} \mathbf{J}_\alpha \mathbf{R}^T, \quad \mathbf{I}_b = \mathbf{R} \mathbf{J}_b, \quad (5.7)$$

$$\mathbf{J}_\rho = \int_A \rho \tilde{\mathbf{X}}^T \tilde{\mathbf{X}} \, dA = \rho \begin{bmatrix} I_{yy} + I_{zz} & 0 & 0 \\ 0 & I_{zz} & -I_{yz} \\ 0 & -I_{yz} & I_{yy} \end{bmatrix}, \quad (5.8)$$

$$\mathbf{J}_{\bar{\omega}} = \int_A \rho \tilde{\mathbf{n}}_{\bar{\omega}}^T \tilde{\mathbf{n}}_{\bar{\omega}} \, dA = \rho \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{\bar{\omega}} & 0 \\ 0 & 0 & I_{\bar{\omega}} \end{bmatrix}, \quad (5.9)$$

$$\mathbf{J}_\alpha = \int_A \rho (\tilde{\mathbf{X}}^T \tilde{\mathbf{n}}_{\bar{\omega}} + \tilde{\mathbf{n}}_{\bar{\omega}}^T \tilde{\mathbf{X}}) \, dA = \rho \begin{bmatrix} 0 & -I_{yc} & -I_{zc} \\ -I_{yc} & 0 & 0 \\ -I_{zc} & 0 & 0 \end{bmatrix}, \quad (5.10)$$

$$\mathbf{J}_b = \int_A \rho \tilde{\mathbf{X}}^T \mathbf{n}_{\bar{\omega}} \, dA = \rho \begin{bmatrix} 0 \\ -I_{zc} \\ I_{yc} \end{bmatrix}, \quad (5.11)$$

$$\mathbf{J}_\alpha = \alpha^2 \mathbf{J}_{\bar{\omega}} + \alpha \mathbf{J}_a, \quad J_{\bar{\omega}} = \rho I_{\bar{\omega}}, \quad (5.12)$$

$$I_{\bar{\omega}} = \int_A \bar{\omega}^2 \, dA = I_\omega + z_c^2 I_{yy} + y_c^2 I_{zz} - 2 y_c z_c I_{yz}, \quad (5.13)$$

$$I_{yc} = \int_A y \bar{\omega} \, dA = -y_c I_{yz} + z_c I_{yy}, \quad (5.14)$$

$$I_{zc} = \int_A z \bar{\omega} \, dA = -y_c I_{zz} + z_c I_{yz}, \quad (5.15)$$

and $I_{yz}, I_{yy}, I_{\omega}$ are defined in (4.100).

In comparison with the kinetic energy for a solid cross-section in Eq. (4.117), the second integral in Eq. (5.6) is the additional kinetic energy due to the warping deformations and the eccentricity of shear center with respect to the centroid.

After several calculations [40], the variation of the total kinetic energy is obtained as

$$\delta K = - \int_{l_0} \left\{ \delta \mathbf{u}_G^T A_\rho \ddot{\mathbf{u}}_G + \delta \mathbf{w}^{wT} \left[\mathbf{I}_\rho^1 \ddot{\mathbf{w}}^w + \mathbf{I}_\rho^2 \dot{\mathbf{w}}^w \right] \right\} dl, \quad (5.16)$$

with

$$\delta \mathbf{w}^w = \begin{bmatrix} \delta \mathbf{w} \\ \delta \alpha \end{bmatrix}, \quad \dot{\mathbf{w}}^w = \begin{bmatrix} \dot{\mathbf{w}} \\ \dot{\alpha} \end{bmatrix}, \quad \ddot{\mathbf{w}}^w = \begin{bmatrix} \ddot{\mathbf{w}} \\ \ddot{\alpha} \end{bmatrix}, \quad (5.17)$$

$$\mathbf{I}_\rho^1 = \begin{bmatrix} (\mathbf{I}_\rho + \mathbf{I}_\alpha) & -\mathbf{I}_b \\ -\mathbf{I}_b^T & J_{\bar{\omega}} \end{bmatrix}, \quad \mathbf{I}_\rho^2 = \begin{bmatrix} \tilde{\mathbf{w}} (\mathbf{I}_\rho + \mathbf{I}_\alpha) & \mathbf{I}_c \dot{\mathbf{w}} \\ -\dot{\mathbf{w}}^T \mathbf{I}_\alpha' & 0 \end{bmatrix}. \quad (5.18)$$

5.2 Beam kinematics

The corotational beam kinematics, adopted here, is similar to the one given in Section 4.1. The only difference is that in order to include warping effects, additional (warping) degrees of freedom α_i ($i = 1, 2$) are introduced at both nodes of the element. The local displacement vector thus becomes

$$\mathbf{d}_l^w = \left[\bar{u} \quad \bar{\boldsymbol{\theta}}_1^T \quad \bar{\boldsymbol{\theta}}_2^T \quad \alpha_1 \quad \alpha_2 \right]^T. \quad (5.19)$$

However, since the warping is in itself a deformational quantity, these additional degrees of freedom remain constant during the sequence of transformations defined in Section 4.1. Hence, referring for instance to Eq. (4.30), the variations $\delta \mathbf{d}_a$ and $\delta \mathbf{d}_g$ are rewritten as

$$\delta \mathbf{d}_a^w = \left[\delta \bar{u} \quad \delta \bar{\mathbf{w}}_1^T \quad \delta \bar{\mathbf{w}}_2^T \quad \delta \alpha_1 \quad \delta \alpha_2 \right]^T, \quad (5.20)$$

$$\delta \mathbf{d}_g^w = \left[\delta \mathbf{u}_1^{gT} \quad \delta \mathbf{w}_1^{gT} \quad \delta \mathbf{u}_2^{gT} \quad \delta \mathbf{w}_2^{gT} \quad \delta \alpha_1 \quad \delta \alpha_2 \right]^T, \quad (5.21)$$

and the corresponding connection is defined by

$$\delta \mathbf{d}_a^w = \mathbf{B}_g^w \delta \mathbf{d}_g^w, \quad \mathbf{B}_g^w = \begin{bmatrix} \mathbf{B}_g & \mathbf{0}_2 \\ \mathbf{0}_2 & \mathbf{I}_2 \end{bmatrix}, \quad (5.22)$$

with $\mathbf{B}_g$ as given in Eq. (4.30) and $\mathbf{I}_2$ denoting a 2x2 identity matrix.

5.3 Local beam kinematic description

The local motion of a beam cross-section from the initial (i.e. rotated but still undeformed) configuration to the current configuration is defined by $[\bar{u}_1 \ \bar{u}_2 \ \bar{u}_3]^T$, the translation of the cross-section centroid G and $\bar{\boldsymbol{\theta}} = [\bar{\theta}_1 \ \bar{\theta}_2 \ \bar{\theta}_3]^T$, the local rotation of the section and α , the warping intensity (see Eq. (5.3)).

Linear interpolation is used for the axial displacement whereas cubic interpolations are used for the transverse displacements and for the axial rotation. Then, due to the particular choice of the local degrees of freedom, one has

$$\begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \\ \bar{\theta}_1 \\ \bar{\theta}_2 \\ \bar{\theta}_3 \\ \alpha \end{bmatrix} = \begin{bmatrix} N_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & N_3 & 0 & 0 & N_4 & 0 & 0 \\ 0 & 0 & -N_3 & 0 & 0 & -N_4 & 0 & 0 & 0 \\ 0 & N_7 & 0 & 0 & N_8 & 0 & 0 & N_3 & N_4 \\ 0 & 0 & N_5 & 0 & 0 & N_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & N_5 & 0 & 0 & N_6 & 0 & 0 \\ 0 & N_9 & 0 & 0 & -N_9 & 0 & 0 & N_5 & N_6 \end{bmatrix} \mathbf{d}_l^w, \quad (5.23)$$

where the local nodal displacement vector $\mathbf{d}_l^w$ is defined by Eq. (5.19).

The expressions of the interpolation functions are given by

$$\begin{aligned} N_1 &= 1 - \frac{x}{l_o}, \quad N_2 = 1 - N_1, \quad N_3 = x \left(1 - \frac{x}{l_o}\right)^2, \\ N_4 &= -\left(1 - \frac{x}{l_o}\right) \frac{x^2}{l_o}, \quad N_5 = \left(1 - \frac{3x}{l_o}\right) \left(1 - \frac{x}{l_o}\right), \quad N_6 = \left(\frac{3x}{l_o} - 2\right) \frac{x}{l_o}, \\ N_7 &= 1 - \frac{3x^2}{l_o^2} + \frac{2x^3}{l_o^3}, \quad N_8 = 1 - N_7, \quad N_9 = \frac{6x^2}{l_o^3} - \frac{6x}{l_o^2}. \end{aligned}$$

Let $\mathbf{u}_l = [0 \ \bar{u}_2 \ \bar{u}_3]^T$ denotes the local transverse displacement vector. From Eq. (5.23), this vector is given by

$$\mathbf{u}_l = \mathbf{P}_1 \begin{bmatrix} \bar{\theta}_1 \\ \bar{\theta}_2 \end{bmatrix}, \quad (5.24)$$

with

$$\mathbf{P}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & N_3 & 0 & 0 & N_4 \\ 0 & -N_3 & 0 & 0 & -N_4 & 0 \end{bmatrix}. \quad (5.25)$$

From Eq. (5.23), the local rotation and the warping degree of freedom are given by

$$\begin{bmatrix} \bar{\boldsymbol{\theta}} \\ \alpha \end{bmatrix} = \begin{bmatrix} \mathbf{P}_2 & \mathbf{P}_3 \\ \mathbf{P}_4 & \mathbf{P}_5 \end{bmatrix} \begin{bmatrix} \bar{\boldsymbol{\theta}}_1 \\ \bar{\boldsymbol{\theta}}_2 \\ \alpha_1 \\ \alpha_2 \end{bmatrix}, \quad (5.26)$$

with

$$\begin{aligned} \mathbf{P}_2 &= \begin{bmatrix} N_7 & 0 & 0 & N_8 & 0 & 0 \\ 0 & N_5 & 0 & 0 & N_6 & 0 \\ 0 & 0 & N_5 & 0 & 0 & N_6 \end{bmatrix}, & \mathbf{P}_3 &= \begin{bmatrix} N_3 & N_4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \\ \mathbf{P}_4 &= \begin{bmatrix} N_9 & 0 & 0 & -N_9 & 0 & 0 \end{bmatrix}, & \mathbf{P}_5 &= \begin{bmatrix} N_5 & N_6 \end{bmatrix}. \end{aligned} \quad (5.27)$$

5.4 Internal force vector and tangent stiffness matrix

Including warping effects, the global internal force vector and tangent stiffness matrix are given by

$$\mathbf{f}_g^w = \mathbf{B}_g^{wT} \mathbf{f}_a^w, \quad \mathbf{K}_g^w = \mathbf{B}_g^{wT} \mathbf{K}_a^w \mathbf{B}_g^w + \mathbf{K}_m^w, \quad \mathbf{K}_m^w = \begin{bmatrix} \mathbf{K}_m & \mathbf{0}_2 \\ \mathbf{0}_2 & \mathbf{0}_2 \end{bmatrix}, \quad (5.28)$$

with $\mathbf{K}_m^w$ as defined in Eq. (4.77). The two other transformations are modified in a similar way.

The local internal force vector $\mathbf{f}_l^w$ and the local tangent stiffness matrix $\mathbf{K}_l^w$ associated with $\delta \mathbf{d}_l^w$ are derived using the shape functions introduced in Section 5.3 and the theory presented in Section 4.4. However, to incorporate the bending shear deformations, the Hermitian shape functions for the transverse displacements are slightly modified as suggested in the Interdependent Interpolation Element (IIE) [51].

Vlasov's assumption is adopted, i.e.

$$\alpha_{,x} = \bar{\boldsymbol{\theta}}_{1,xx}, \quad (5.29)$$

which, since the nonlinear terms in the expression of the curvatures are neglected, is the direct consequence of Eq. (4.106). A low order of geometrical nonlinearity is introduced through a shallow arch strain description and the Wagner term.

The element was developed using Eqs. (4.95), (4.98) and (4.110). The Maple codes for $\mathbf{f}_l^w$ and $\mathbf{K}_l^w$ were given in [40].

5.5 Inertia force vector and tangent dynamic matrix

The inertia force vector is then derived from the following relation

$$\delta K = -\mathbf{f}_k^{wT} \delta \mathbf{d}_g^w. \quad (5.30)$$

Translational displacement variables

The variation of the translational displacement of G is not affected by the warping deformations, therefore the translational displacement variables are calculated in the same manner as presented in Section 4.5. The expression is given as follows

$$\delta \mathbf{u} = \mathbf{R}_r \mathbf{H}_1 \mathbf{E}^T \delta \mathbf{d}_g, \quad (5.31)$$

where

$$\mathbf{H}_1 = \mathbf{N} + \mathbf{P}_1 \mathbf{P} - \tilde{\mathbf{u}}_l \mathbf{G}^T. \quad (5.32)$$

Finite rotations and warping variables

Combining Eqs. (4.28) and (5.26) with the approximation defined in Eq. (4.122), the local spatial spin and warping variables are calculated as follows

$$\delta \bar{\mathbf{w}} = \begin{bmatrix} \mathbf{P}_2 & \mathbf{P}_3 \end{bmatrix} \begin{bmatrix} \mathbf{P} \mathbf{E}^T & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix} \delta \mathbf{d}_g^w = \begin{bmatrix} \mathbf{P}_2 \mathbf{P} & \mathbf{P}_3 \end{bmatrix} \mathbf{E}^{wT} \delta \mathbf{d}_g^w, \quad (5.33)$$

$$\delta \alpha = \begin{bmatrix} \mathbf{P}_4 & \mathbf{P}_5 \end{bmatrix} \begin{bmatrix} \mathbf{P} \mathbf{E}^T & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix} \delta \mathbf{d}_g^w = \mathbf{H}_3 \mathbf{E}^{wT} \delta \mathbf{d}_g^w, \quad (5.34)$$

where

$$\mathbf{E}^w = \begin{bmatrix} \mathbf{E} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \quad \mathbf{H}_3 = \begin{bmatrix} \mathbf{P}_4 \mathbf{P} & \mathbf{P}_5 \end{bmatrix}. \quad (5.35)$$

Combining Eq. (5.33) with Eq. (4.25), and using Eq. (4.28), the spatial spin variables, associated to the global rotation of a cross-section, are evaluated using

$$\delta \mathbf{w} = \mathbf{R}_r \mathbf{H}_2 \mathbf{E}^{wT} \delta \mathbf{d}_g^w, \quad (5.36)$$

with

$$\mathbf{H}_2 = \begin{bmatrix} \mathbf{P}_2 \mathbf{P} + \mathbf{G}^T & \mathbf{P}_3 \end{bmatrix}. \quad (5.37)$$

Introducing Eqs. (5.34) and (5.36) into Eq. (5.17), one obtains

$$\delta \mathbf{w}^w = \begin{bmatrix} \mathbf{R}_r & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{H}_2 \\ \mathbf{H}_3 \end{bmatrix} \mathbf{E}^{wT} \delta \mathbf{d}_g^w = \mathbf{R}_r^w \mathbf{H}_2^w \mathbf{E}^{wT} \delta \mathbf{d}_g^w, \quad (5.38)$$

Inertia force vector and tangent dynamic matrix

By inserting the expressions of $\delta \mathbf{u}$ and $\delta \mathbf{w}^w$ given in Eqs. (5.31) and (5.38) into Eq. (5.16), the inertia force vector is obtained as

$$\mathbf{f}_k^w = \mathbf{E}^w \left\{ \begin{bmatrix} \int_{l_o} \mathbf{H}_1^T \mathbf{R}_r^T A_\rho \ddot{\mathbf{u}} dl \\ \mathbf{0} \end{bmatrix} + \int_{l_o} \mathbf{H}_2^T \mathbf{R}_r^T \left[\mathbf{I}_\rho^1 \ddot{\mathbf{w}}^w + \mathbf{I}_\rho^2 \dot{\mathbf{w}}^w \right] dl \right\}. \quad (5.39)$$

The iterative scheme of the present formulation is implemented with the following approximate linearization

$$\Delta \mathbf{f}_k^w \approx \mathbf{M} \Delta \ddot{\mathbf{d}}_g^w + \mathbf{C}_k \Delta \dot{\mathbf{d}}_g^w. \quad (5.40)$$

The expression of the mass matrix $\mathbf{M}$ is given by

$$\begin{aligned} \mathbf{M} &= \mathbf{E}^w \left\{ \begin{bmatrix} \int_{l_o} \mathbf{H}_1^T A_\rho \mathbf{H}_1 dl & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \int_{l_o} \mathbf{H}_2^T \mathbf{R}_r^T \mathbf{I}_\rho^1 \mathbf{R}_r^w \mathbf{H}_2^w dl \right\} \mathbf{E}^{wT} \\ &= \mathbf{E}^w \mathbf{M}^e \mathbf{E}^{wT}. \end{aligned} \quad (5.41)$$

And the gyroscopic matrix $\mathbf{C}_k$ is calculated from

$$\begin{aligned} \mathbf{C}_k &= \mathbf{E}^w \left\{ \begin{bmatrix} \int_{l_o} \mathbf{H}_1^T A_\rho (\mathbf{C}_1 + \mathbf{C}_3) dl & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} + \int_{l_o} \mathbf{H}_2^T \mathbf{R}_r^T \mathbf{I}_\rho^1 \mathbf{R}_r^w (\mathbf{C}_2 + \mathbf{C}_4) dl \right. \\ &\quad \left. + \int_{l_o} \mathbf{H}_2^T \mathbf{R}_r^T (\mathbf{I}_\rho^2 + \mathbf{I}_\rho^3) \mathbf{R}_r^w \mathbf{H}_2^w dl \right\} \mathbf{E}^{wT} = \mathbf{E}^w \mathbf{C}_k^e \mathbf{E}^{wT}, \end{aligned} \quad (5.42)$$

where

$$\mathbf{I}_\rho^3 = \begin{bmatrix} \widetilde{(\mathbf{I}_\rho + \mathbf{I}_\alpha)} \dot{\mathbf{w}} + \mathbf{I}_c \dot{\alpha} & \mathbf{0} \\ -\dot{\mathbf{w}}^T \mathbf{I}_\alpha'^T & 0 \end{bmatrix}. \quad (5.43)$$

The warping deformations and the eccentricity of the shear center have been taken into account in deriving the inertial terms, see Eq. (5.4). Consequently, several additional terms were introduced in the expressions of the inertia force vector and the tangent (mass and gyroscopic) matrices. The importance of these terms were discussed in [40] through several numerical examples. It was found, as expected, that the warping deformations in the dynamic terms had a negligible contribution to the response and could be omitted. This is done by setting $I_\omega = 0$ in Eq. (5.13).

The eccentricity of the shear center with respect to the centroid generates extra dynamic terms whose importance was also investigated. It appeared that for slender beams, these terms did not affect the results and can be therefore neglected. But it also showed that they may have a significant contribution for short beams.

If $I_\omega = y_c = z_c = 0$ is adopted, then the expression of the inertia force vector in Eq. (5.39) can be simplified as

$$\mathbf{f}_k^w = \mathbf{E}^w \left\{ \begin{bmatrix} \int_{l_0} \mathbf{H}_1^T \mathbf{R}_r^T A_\rho \ddot{\mathbf{u}} dl \\ \mathbf{0} \end{bmatrix} + \int_{l_0} \mathbf{H}_2^T \mathbf{R}_r^T \left[\mathbf{I}_\rho \ddot{\mathbf{w}} + \tilde{\mathbf{w}} \mathbf{I}_\rho \dot{\mathbf{w}} \right] dl \right\}, \quad (5.44)$$

and similar simplifications apply also for the expressions of the mass and gyroscopic matrices.

5.6 Numerical example - Short cantilever beam with an U cross-section

In [40], five numerical examples of beams with thin-walled cross-sections were solved using the proposed formulation. The predictions of the new formulation was compared against 3D-solid analyses performed with the commercial finite element program Abaqus. The results have demonstrated the accuracy of the formulation. In this chapter, the last example of the five is reproduced. The aim of the this example is not only to assess the performance of the present formulation, but also to highlight the influence of the eccentricity of the shear center on the inertia terms. In fact, for the first, second and fourth examples in [40], the latter did not affect the numerical results and could be neglected by replacing $y_c = z_c = 0$ in Eqs. (5.39), (5.41) and (5.42). But, this example will show a case when the contribution of the eccentricity of the shear center is important.

The cantilever beam with an U cross-section, see Fig. 5.2, was analysed. The eccentricity of the shear center is given by $z_c = -0.138$ m. The beam is subjected to two out-of-plane time-varying concentrated loads, one at the middle and one at the right end. The time-varying load applied at point A is $F_z^A = -4F$. The time-varying load applied at point B is $F_z^B = 2F$. The time evolution of F is given in Fig. 5.3.

The following material properties were used: $E = 210$ GPa, $\nu = 0.33$ and $\rho = 7850$ kg/m³. All the dimensions in the figures were in meter.

Two different corotational beam models with 40 elements were employed. In the first one, the warping deformations and the eccentricity of the shear center were incorporated in the inertia terms while in the second one they were neglected.

The solid mesh consisted of $16 \cdot 60 = 960$ isoparametric 20 node elements. For the 3D-solid model, four connectors were added at sections C1-C4 along the beam (see Fig. 5.4) in order to avoid in-plane deformations of the flanges. The connector between 2 nodes maintained the distance of these nodes during the analysis.

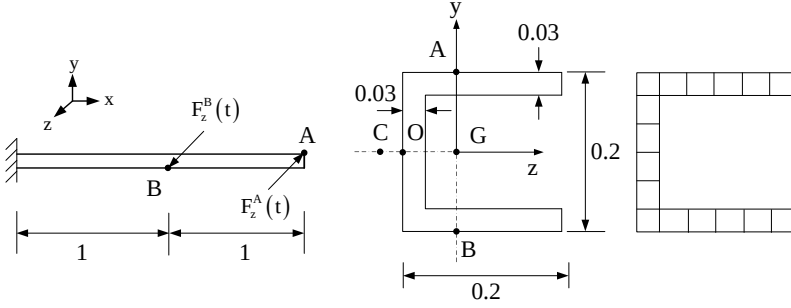


Fig. 5.2: Cantilever beam with U cross-section : geometrical data.

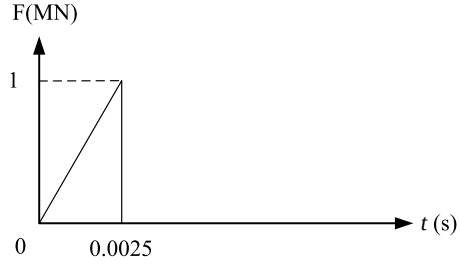


Fig. 5.3: Cantilever beam with U cross-section - Loading history.

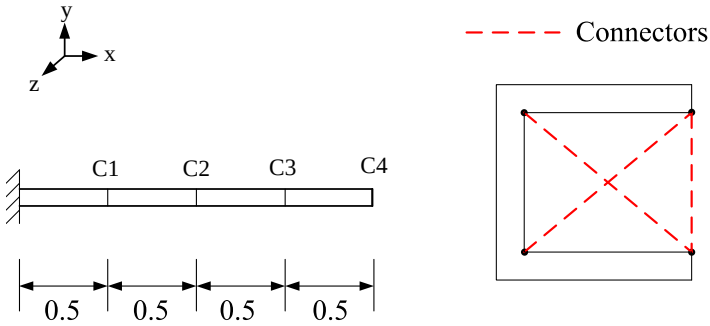


Fig. 5.4: Cantilever beam with U cross-section : Connectors in the 3D-solid model.

The first ten linear natural frequencies of the cantilever were calculated with all the models.

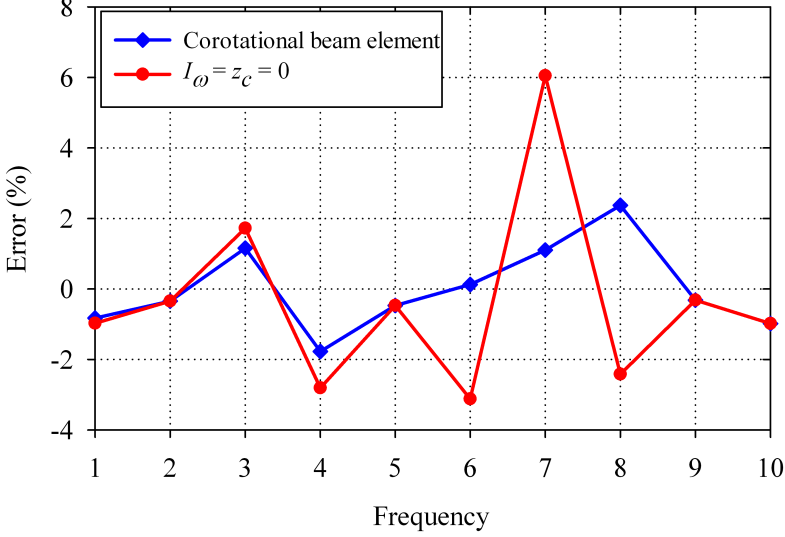


Fig. 5.5: Cantilever beam with U cross-section - Natural frequency analysis.

For the beam models, a linearized version of the corotational formulation was used. The difference between the i^{th} natural frequency given by a beam model and by the 3D-solid model was calculated as follow

$$Error^i = \left(\frac{Fre_{Beam}^i}{Fre_{Solid}^i} - 1 \right) \cdot 100\%. \quad (5.45)$$

The results, depicted in Fig. 5.5, showed that, for the pure bending modes 2th, 5th, 10th and the axial mode 9th, the warping deformations and the eccentricity of the shear center did not effect the frequencies. However, for the torsion-bending modes, the frequencies were better estimated by the beam element which incorporated the warping deformations and the eccentricity of the shear center. The differences between this beam analysis and the 3D-solid one remained smaller than 2.5%. But, when $I_{\omega} = z_c = 0$, the differences could reach 6%.

The nonlinear behavior of the cantilever due to the time-varying loads were investigated. All analyzes were performed with a time step $\Delta t = 5 \cdot 10^{-5}$ s. Regarding the time integration method, the HHT α method with $\alpha = -0.05$ was used. At the beginning of each time step, the predictor Pred. 4 (see Section 3.2) was adopted. The time evolution of the displacement u_y of point O at the right end was depicted in Fig. 5.6. A very good agreement between the corotational beam and Abaqus 3D-solid analysis was obtained. However,

when $I_\omega = z_c = 0$, large discrepancies between the beam and 3D-solid analysis could be noted.

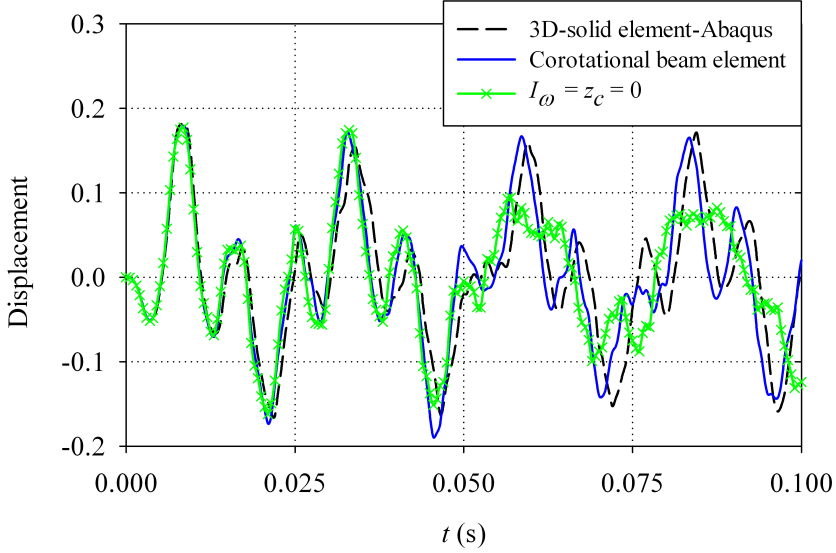


Fig. 5.6: Cantilever beam with U cross-section - Time evolution of the displacement u_y of point O.

This example shows that the proposed element can be used to analyse the nonlinear dynamic behavior of beams with arbitrary cross-sections and also that the additional dynamic terms due to the eccentricity of the shear center cannot be neglected. As a matter of fact, the numerical results seem to be improved if the eccentricity of the shear center is considered.

Chapter 6

Conclusions and future research

6.1 Conclusions

The main objective of this research was to provide efficient beam formulations for nonlinear dynamics of flexible structures using the corotational method. Beam formulations in both 2D and 3D (with solid and arbitrary thin-walled cross-section) were considered.

2D beam formulation

A consistent dynamic beam formulation for the corotational approach in 2D was derived. The same corotational kinematic description was adopted for both static and dynamic parts. Moreover, in order to take into account the inertia effect due to the local transverse displacements, the local cubic interpolations were not only adopted to develop expressions of the elastic terms but also to derive the inertia terms.

This new formulation was compared with two classic formulations using constant Timoshenko and constant lumped mass matrices. Several numerical examples showed that the new formulation was more efficient than the two classic ones. Indeed, all results indicated that the new formulation required more computational time but allowed to reduce significantly the number of elements. This advantage is due to a better representation of the local displacements in the inertia terms.

Finite rotations in dynamics

The treatment of the finite rotations is one of the most important issues in the development of 3D beam elements. In the paper [38], the rotation tensor, the incremental rotational vector and the incremental Euler parameters were adopted to parameterize the finite rotations. Four dynamic formulations based on these parameterizations were compared. The first

three formulations were taken from the literature [11, 30, 55]. The last one was new. In these formulations, the corotational method was used to develop expressions of the internal forces and the tangent stiffness matrices, while the dynamic terms were formulated into a total Lagrangian context. The similarities and differences between the formulations in theoretical derivations and implementations were pointed out. Many numerical examples were implemented in order to assess numerical accuracy and computational efficiency of the formulations. Beside several predictors for iterative procedure and various possibilities to simplify the tangent dynamic matrix were tested.

The numerical results showed that

- For the practical implementation, the new formulation and the one proposed by Cardona and Geradin [11] were the simplest ones: the same standard Newmark method was used for both the displacements and the rotations. No quantities needed to be saved at each Gauss point.
- For all the formulations, the predictor proposed by Crisfield in [13] was the best alternative. This predictor for the general case of a non constant tangent inertia matrix was given in detail in the second paper [38].
- For each formulation, the choice of the tangent inertia matrix affects significantly the computational efficiency. For the first two formulations, the best alternative was to take the exact tangent inertia matrix. For the last two formulations, it was better to take only the mass and gyroscopic matrices.
- All the formulations gave the same numerical results but the new formulation was slightly more efficient than the other ones regarding the CPU time.

3D beam formulation with solid cross-section

A consistent 3D corotational beam element for nonlinear dynamics of flexible structures was presented. Compared with the other formulations found in the literature, the novelty of the formulation is that the corotational kinematics was used to derive both internal and inertia terms. Besides, local cubic interpolation functions were used. The inertia force vector and the tangent dynamic matrix were derived by introducing an approximation with regard to the local rotations. To enhance the efficiency of the iterative procedure, an approximative tangent dynamic matrix was adopted.

Four numerical examples were considered to compare the new formulation with two other approaches. The first approach, called as “linear corotational approach”, was derived from the new formulation by replacing cubic interpolations with linear ones. The second approach was the classic total Lagrangian proposed by Simo and Vu-Quoc [53–55]. This approach can be implemented with different number of nodes. To ensure a meaningful comparison of both formulations, Simo and Vu-Quoc’s approach was implemented with two nodes which correspond to a linear interpolations for all global variables.

Based on these examples, the following conclusions were drawn

- All formulations gave the same results with fine meshes.
- As expected, with coarse meshes, Simo and Vu-Quoc's two-noded formulation and the linear corotational one gave less accurate results compared to the cubic corotational approach which provided very accurate results. This demonstrated the importance of an accurate description of the deformations in the dynamic terms, which was not possible with linear interpolations.
- The linear corotational approach was faster than the cubic one. This was expected since several inertia terms disappeared when local linear interpolations were adopted. Moreover, in the linear corotational approach, the inertia force vector and the tangent dynamic matrix were integrated using two Gauss points, whereas the dynamic terms of the cubic corotational one were integrated using three Gauss points.
- As it was more involved, the cubic corotational formulation was slightly slower than linear formulations (around 12%). However, certain cases could arise where the desired accuracy could be achieved via the cubic formulation faster than the linear formulation of Simo and Vu-Quoc.

One could conclude that the new dynamic corotational formulation based on the local cubic interpolations provided sufficient accuracy with relatively low number of elements as compared to linear formulations. With the same number of elements, about the same computational time was required (this could differ a little from one example to another). However, with the present formulation, accurate results could be obtained with a smaller number of elements.

3D beam formulation with arbitrary thin-walled open cross-section

The previous 3D corotational beam element is extended for the nonlinear dynamics of structures with thin-walled cross-section by introducing the warping deformations and the eccentricity of the shear center. This led to additional terms in the expressions of the inertia force vector and the tangent (mass and gyroscopic) dynamic matrices. This element had seven degrees of freedom at each node.

Five 3D beam problems, with different thin-walled cross-sections and load pattern, were analysed. The predictions of the proposed formulation were compared against the numerical results obtained with both Abaqus beam element and 3D-solid analyses. For all the examples, a very good agreement between the results obtained with the corotational beam element and the 3D-solid element was obtained. Besides, the numerical results also showed that the warping deformations had a negligible influence in the dynamic terms and could be omitted. However, the additional dynamic terms due to the shear center eccentricity could not always be neglected. In fact, it seemed that for slender beams, the same

numerical results were obtained with or without these terms. However, for short beams, the numerical results seemed to be improved if the shear center eccentricity was considered.

6.2 Future research

From the work presented in this thesis, some interesting topics for future research are suggested.

Energy conserving method

The HHT α method was used to solve the nonlinear equation of motion in this work. This method introduces a small numerical damping and limits the influence of high frequencies. However, the numerical results can be affected for long time analyses due to the dissipation of the total energy. It should be noted that the main objective of this work is to compare different beam elements, thus this disadvantage had no influence to the conclusions.

For beam structures, more robust alternatives to the HHT α method have been proposed in the literature [14, 29]. It would be interesting to implement the new beam elements with an energy conserving method.

Initially curved beam element

All the formulations in this work were developed in the context of two-noded straight beam elements. For beam structures with curved geometry, it is always possible to catch their behaviors by using a large number of straight elements. However, in order to reduce the number of elements, a curved beam element can be used. Hence, an interesting topic for a further research is to extend the elements developed in this thesis by introducing initial curvature.

Comparative study with a total Lagrangian quadratic beam element

As showed in [39], to obtain accuracy results, the new dynamic corotational formulation, based on the local cubic interpolations, required a smaller number of elements than the one of Simo and Vu-Quoc, based on linear interpolations. Although, it must be emphasized that the above conclusion was based on a comparative study of two-noded elements. In the context of the total Lagrangian approach, efficient three-noded elements with quadratic shape functions can be developed, see for instance the formulation for static analysis proposed in [27, 31]. Therefore, it should be interesting for dynamic problems to compare the new corotational formulation and a total Lagrangian quadratic element.

New corotational shell finite elements for dynamics

Several corotational shell elements has been developed by Pacoste [47] and Battini [6]. All of these shell elements concern static analysis. Therefore, it would be interesting to extend the present work to the nonlinear dynamics of corotational shells.

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